

The Unintended Consequences of Clients' Active Participation in Expert Services: Theory and Experiment

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Abstract

In expert services, such as medical services, clients often lack specialized expertise to diagnose their problems. Yet it is still common for them to participate in decision-making by forming their own diagnoses, potentially leading to noncompliance with experts' advice. This study investigates how clients' active participation affects experts' investment in diagnostic precision. In a theory-driven laboratory experiment, clients consult experts to diagnose an unknown problem. Clients are either *passive*, always following the expert's recommendation, or *active*, with the option to overrule the expert. Experts choose how much effort to invest in improving diagnostic precision. The results show that allowing clients to overrule experts reduces experts' investment in diagnostic precision. Moreover, this effect is stronger when experts' incentives are more closely aligned with clients' well-being. These findings highlight an unintended consequence of client participation in expert services: giving clients greater decision-making authority may discourage experts from investing in more accurate diagnoses.

Keywords: Expert services, Client Noncompliance, Lab experiments

JEL Codes: C91, D82, I11

1 Introduction

Expert service differs from other services in that clients lack a clear understanding of their needs: they rely on experts’ specialized skills to diagnose problems and provide treatments. Across various domains such as healthcare, financial consulting, and legal consulting, it is common for clients, despite not having the same level of expertise as the professionals, to participate in problem diagnosis and solution choices. The active participation of clients in expert services raises questions about how experts respond when their authority is questioned, or in other words, when clients appear not fully compliant.

Consider medical service as an example: although the patient-doctor relationship has traditionally been viewed through a top-down paternalist model ([Emanuel and Emanuel, 1992](#)), nowadays more and more patients engage in medical decision-making, partly driven by the democratization of information through the internet ([Tan and Goonawardene, 2017](#)). Using a nationwide survey on internet usage, [Fox and Duggan \(2013\)](#) reported that around 35% of U.S. adult internet users used the internet as a diagnostic tool, with 18% of them experiencing self-diagnoses conflicting with doctors’ assessments. The low-cost access to health information, though not necessarily accurate, empowers patients to take a more active role in their healthcare decisions.

However, it remains unclear how clients’ active participation influences experts’ behaviors. In the medical context, although doctors are generally motivated to improve patients’ health, active client involvement may be perceived as a challenge to their professional judgment, potentially reducing their willingness to exert additional effort to achieve a more accurate diagnosis. Some empirical evidence is consistent with this concern. In a national survey in the US, 59% of doctors reported “patient’s request” as the main reason for over-treatment ([Lyu et al., 2017](#)). Moreover, physician burnout is positively associated with a higher frequency of difficult patient encounters ([An et al., 2013](#)), although the causal direction remains unclear. Due to the challenges of empirical identification, there is a lack of

direct causal evidence on how clients’ active participation (or the possibility of noncompliance) affect experts’ diagnostic and treatment efforts.

In this study, I discuss the influence of clients’ active participation on experts’ investment of effort in the diagnosis. In my theory-driven lab experiment, I introduce two players: a client and an expert, where the client consults the expert to diagnose an unknown problem and recommend a treatment. Following [Balafoutas et al. \(2020\)](#), the expert decides the investment in improving the diagnostic precision, which determines the probability of obtaining a correct diagnosis. This setup captures the fact that, in real-life scenarios, experts’ diagnosis may be associated with uncertainty ([Beresford, 1991](#)). Clients, though lacking expertise in problem diagnosis, have an inaccurate self-diagnosis which may influence their decisions on whether to follow or to go against their experts’ recommendations.

Building on this basic setup, the experiment varies three key elements. First, to understand the influence of clients’ active participation on experts’ investment in diagnostic precision, within each experimental session, I introduce two types of clients, “passive” and “active”. Passive clients always accept experts’ diagnoses and treatment recommendations, while active clients have an option to go against experts. Second, rather than relying on subjects’ inherent social preferences, I introduce an “alignment parameter” to exogenously vary the expert’s incentive to solve the client’s problem. This parameter captures the degree of incentive alignment between experts and clients. Third, this design introduces two treatment conditions, “Observable” and “Concealed” conditions, varying in whether the expert’s diagnostic precision is observable to the client.

The experimental results show that experts treat active and passive clients differently only when diagnostic precision is not observable to the client. When diagnostic precision is observable, experts invest equally in active and passive clients, regardless of their incentive alignment. In contrast, when diagnostic precision is concealed, experts invest less in diagnostic precision for active clients than for passive clients. More importantly, this investment gap widens as experts’ incentive alignment with clients increases. In other words, experts with

stronger incentives to solve clients' problems exhibit a larger reduction of diagnostic precision on active clients when they have an option to overrule their recommendations. Consequently, giving clients an option to overrule experts' recommendation lowers the probability that their problems are successfully solved.

Applied to medical services, this finding suggests that patients' active participation in treatment decisions may, under certain conditions, lead to worse health outcomes. When patients cannot directly assess the accuracy of doctors' diagnoses, doctors may perceive patients' participation in decision-making as potential noncompliance. Doctors who are more strongly motivated to improve patients' health may therefore anticipate that their diagnostic effort is more likely to be wasted and, as a result, invest less effort in the diagnosis. Consequently, compared with patients who always fully follow doctors, patients who are expected to challenge doctors receive less diagnostic effort and face a lower probability of receiving the correct treatment.

This study contributes to the literature on expert services by examining how clients' active participation affects experts' investment in diagnostic accuracy. The experimental results show that when experts are motivated to solve clients' problems but cannot credibly disclose their diagnostic precision, clients' ability to overrule expert recommendations discourages experts from investing in more accurate diagnoses. Identifying this causal effect is difficult using observational data because clients' participation and experts' effort are jointly determined in real-world settings.

The study also highlights the role of incentive alignment in shaping experts' responses to client behavior. Counterintuitively, experts whose incentives are more closely aligned with clients' interests exhibit a larger reduction in diagnostic effort when clients are less likely to comply with their recommendations. In the context of healthcare, incentive alignment can be interpreted as physicians' concern for patients' well-being. More broadly, different degrees of incentive alignment may arise from organizational incentives or contractual arrangements that tie experts' payoffs more closely to clients' outcomes. The findings suggest that in

expert-service settings where experts' welfare depends more directly on successfully solving clients' problems, experts are more responsive to clients' expected trust and compliance.

2 Related Literature

This study is closely related to the expert service literature. Previous studies on clients' active participation in expert services and expert's other-regarding preferences are discussed in this subsection. In addition, under the medical context, this study complements the empirical studies on healthcare provider behaviors and adds to the literature on experimental health economics.

2.1 Client Participation, Diagnostic Uncertainty, and Expert Social Preferences in Expert Services

Expert services, where experts possess more knowledge about clients' needs, are first studied by [Darby and Karni \(1973\)](#). In these so-called credence good markets, such as healthcare and car repair services, [Dulleck and Kerschbamer \(2006\)](#) predicts that when clients fully commit to experts' recommended treatments, the information asymmetry can result in market failure. Subsequent research has explored how clients' participation affects the market equilibrium, focusing on behaviors such as rejecting expert advice, seeking second opinions, and conducting personal research on diagnoses. These studies offer mixed findings on the impact of client participation. [Mimra et al. \(2016\)](#) demonstrate through a laboratory experiment that access to costly second opinions significantly reduces experts' overtreatment. As pointed out by [Balafoutas and Kerschbamer \(2020\)](#), improved access to information enhances client autonomy and reduces expert dishonesty. Conversely, other studies highlight potential drawbacks of client participation in decision-making. For example, [Fong et al. \(2014\)](#) finds that if clients do not commit to experts' recommendations, the market can end up inefficient even if service quality is verifiable. [Dulleck and Kerschbamer \(2009\)](#) examine how clients'

free-riding, where clients get a diagnosis from experts but then switch to cheaper providers for treatment, affects expert diagnoses. They combine the ideas of expert diagnostic uncertainty and consumer non-commitment and find that this free-riding discourages experts from putting in more effort, resulting in undertreatment.¹ My study extends these findings by showing how clients’ active participation can discourage experts from providing better services, especially when experts are highly motivated to solve clients’ problems.

In addition, there are two important settings discussed in the expert service literature, which are closely related to this study: diagnostic uncertainty, and expert’s other-regarding preferences. In recent years, there have been a few studies discussing the role of diagnostic uncertainty, as exogenously determined by experts’ ability or endogenously determined by experts’ investment of effort. [Hilger \(2016\)](#) extends the model developed by [Dulleck and Kerschbamer \(2006\)](#), demonstrating that when the diagnostic cost is not observable, mistreatment always exists in equilibrium. [Liu et al. \(2020\)](#) investigates the market equilibrium where experts are heterogeneous in their diagnostic ability. Their theoretical results imply that under certain settings, a higher proportion of high-ability experts in the market can reduce market efficiency. [Balafoutas et al. \(2020\)](#) discuss diagnostic uncertainty in the context of insurance coverage, finding that insurance reduces experts’ investment in the diagnosis and lowers the diagnostic precision. The theoretical framework of my study borrows the setup from [Balafoutas et al. \(2020\)](#), assuming that experts have to exert costly effort to improve the precision of the diagnosis.

Regarding experts’ other-regarding preferences, [Kerschbamer et al. \(2017\)](#) designed an experiment to clearly distinguish experts’ heterogeneous social preferences, showing that social preferences better explain experts’ behaviors than the standard assumption of profit-maximizing preferences. In the experiment by [Balafoutas et al. \(2020\)](#), experts’ social pref-

¹Clients’ participation in decision-making can also occur in earlier stages of the service. For example, [Schulte and Felgenhauer \(2017\)](#) study the impact of clients’ participation in the pre-selection of investment projects. They find that if a client pre-selects a project for an expert to evaluate, the expert will know that the client favors this project and will be biased towards recommending executing this project. As a result, clients’ involvement in the selection of projects increases the risk of investing in a bad project.

erences also play an important role in improving diagnostic precision. In my theoretical framework, I use the term “incentive alignment” to generalize the experts’ other-regarding preferences into the alignment of motivation on problem-solving between experts and clients. I add to the expert service literature by showing that incentive alignment between experts and clients plays a vital role in experts’ perspectives of clients’ active participation. Intuitively, experts who are more concerned about clients’ well-being are more sensitive to clients’ compliance.

2.2 Healthcare Services: Impacts of Patient Characteristics on Doctor Provision

One of the primary applications of the findings from this study is in healthcare services. There is a small but growing body of literature using laboratory experiments to investigate healthcare-related topics, such as the design of incentive schemes ([Brosig-Koch et al., 2016, 2017](#); [Hennig-Schmidt et al., 2011](#)). As noted by [Galizzi and Wiesen \(2018\)](#), there is increasing acceptance of the laboratory approach in health economics, alongside the application of behavioral economics to health research. My experimental findings complement empirical studies on patient-doctor relationships. Existing empirical studies on this topic indicate that doctors do not treat patients uniformly. Their medical treatments vary based on patients’ education levels ([Brekke et al., 2018](#)), socio-economic status ([Banuri et al., 2018](#); [Gottschalk et al., 2020](#)), ethnicity ([Alsan et al., 2019](#)), gender ([Schulman et al., 1999](#)), etc. My study adds to this discussion by suggesting that patients’ active involvement and potential noncompliance can also affect doctors’ service quality. Specifically, doctors with a stronger motivation to cure illness are more sensitive to whether their advice will be considered seriously. This implication, to my best knowledge, has not been fully explored in the literature.

3 Theoretical Framework: Expert-Client Interaction

3.1 Basic Setup

I consider an economy with one client and one expert. The client faces an unknown problem, z , and must visit an expert for diagnosis and treatment to solve the problem. To simplify the discussion, I assume there are two problems $z \in \{0, 1\}$, with an equal ex-ante probability of 0.5 for each, meaning these two problems are equally likely to occur. Additionally, there are two treatments $T \in \{0, 1\}$ that can solve each of these problems: treatment $T = 1$ solves $z = 1$, and treatment $T = 0$ solves $z = 0$.

The client receives a private signal $s^c \in \{0, 1\}$ about the problem, with the precision $Pr(s^c = z) = q$, where $q \in [0.5, 1]$. This private signal represents the client's self-diagnosis of the unknown problem, similar to real-life situations where clients may use online searches to diagnose issues, which are not always accurate. The expert receives a diagnostic signal $s^x \in \{0, 1\}$ that correctly identifies the problem with a probability E , i.e. $Pr(s^x = z) = E$. In line with [Balafoutas et al. \(2020\)](#), I assume that the expert can freely choose the level of diagnostic precision $E \in [0.5, 1]$, which incurs an effort cost $g(E) = k(E - 0.5)^2$, where $k > 0$. I further assume that the expert cannot provide a treatment that is inconsistent with the diagnostic result: after the diagnosis, the diagnostic result is automatically converted to a treatment recommendation $T^x \in \{0, 1\}$, with $T^x = s^x$. This setting mirrors a scenario where increased effort by an expert leads to more precise diagnostic results.² After getting a diagnosis, the expert follows a guideline on giving treatments, without room to manipulate the interpretation of the diagnostic result or to purposely mislead clients into another treatment. This case is typically true in healthcare services where doctors have to follow

²When $E = 0.5$, it means that the expert's diagnosis is a 50/50 random draw which does not convey any useful information about the problem. One may be concerned that in real life, experts usually do not invest so little in the diagnosis. I believe that this simplification is worthwhile, because in the experiment, what I am focusing on is the change in the expert's diagnostic precision between treatments rather than the absolute level of diagnostic precision. Future research could introduce punishment for malpractice.

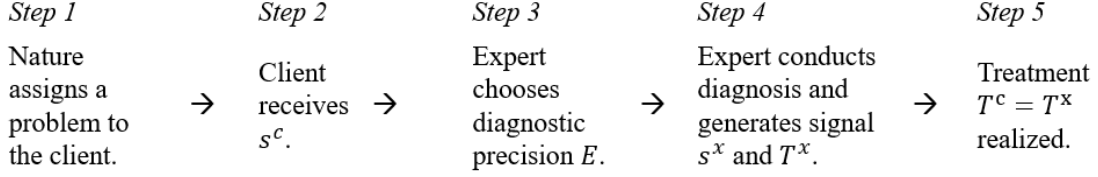


Figure 1: Game Procedure with Passive Client

protocols in treating patients based on diagnostic results.

To simplify the discussion, I assume that the client's goal is to solve the problem. Therefore, the client's utility U^c is determined solely by whether the treatment that the client receives, namely T^c , is the desired treatment: $U^c = H$ if $T^c = z$, and $= L$ if $T^c \neq z$ (assuming H and L are constants, and $H > L$). Following Balafoutas et al. (2020) and Liu et al. (2020), I assume that the expert cares about the client's problem-solving, controlled by an incentive alignment parameter $\gamma \in [0, 1]$. The expert is under a capitation payment system, so he receives a lump sum M for serving one client. Thus, the expert's utility U^x is defined in the following way: $U^x = M + \gamma(U^c - L) - g(E)$. Notice that the term $U^c - L$ equals $H - L$ if the problem is solved and 0 if the problem is not solved. Therefore, the incentive alignment γ determines the extra utility that the expert receives from improving the client's well-being. In other words, this alignment parameter γ captures the expert's concerns about the client's well-being. One could interpret this alignment in various ways, for example, the expert's inherent altruism, the expert's concerns about their reputation, the expert's fear of malpractice, etc.

I consider two types of clients, *passive* clients, and *active* clients. A passive client always accepts the expert's recommended treatment. In the context of healthcare, this setup mirrors the traditional paternalistic model of patient-doctor relationships, where doctors are the primary decision-makers and patients fully comply with doctors (Emanuel and Emanuel, 1992). An active client has the option to actively participate in treatment selection. After receiving the diagnostic result and recommendation of treatment, they can freely go against their experts by choosing the opposite treatment.

In the rest of this section, I will discuss the interaction between the client and the expert in three cases. The first case (“passive client”) is a benchmark case, where the expert interacts with a passive client. Then I further investigate two cases with active clients. In Case 1 (“active client & observable precision”), the client directly observes the expert’s diagnostic precision and determines whether to follow the expert’s recommendation. In Case 2 (“active client & concealed precision”), the client does not observe the expert’s diagnostic precision but is still able to overrule the expert’s recommendation. In Case 2, I assume that the client forms a belief about the expert’s diagnostic precision, with the expectation of the belief denoted as $\hat{E}$. By comparing the passive-client case with the two active-client cases, I will be able to investigate how clients’ active participation changes experts’ behaviors, and how the observability of diagnostic precision plays a role in this relationship.

3.2 Benchmark: Passive Client

The benchmark case focuses on passive clients. Figure 1 shows the game procedure. First, the nature assigns an unknown problem to the client (Step 1). Next, the client receives a private signal s^c about the problem (Step 2), and the expert determines the effort level E (Step 3). After investing the diagnostic effort, a diagnostic result s^x is drawn which identifies the problem with probability E . The diagnostic result s^x will be converted into a treatment T^c for the client (Step 4). The passive client automatically accepts the treatment and both parties’ utilities are realized.³

Proposition 1 *When the client is passive, the expert’s optimal choice of diagnostic precision is: $E^{passive} = 0.5 + \gamma(H - L)/2k$.*

³Given that the client is passive, whether the client receives a private signal or observes the expert’s investment in diagnosis does not affect the equilibrium. However, I still include these steps in Figure 1 to keep the game procedure consistent with the experimental design, which will be explained in the next section.

Proof. The expected utility for the expert is:

$$E(U^x) = M + \gamma(EH + (1 - E)L) - k(E - 0.5)^2 \quad (1)$$

Take the first-order derivatives with respect to E , then the solution to the optimal precision is obvious. ■

Proposition 1 indicates that when the expert knows that the client is fully compliant with the expert's recommendation, the expert's investment of efforts in the diagnostic precision increases with the alignment parameter γ . Notably, when $\gamma = 0$, indicating no motivation to improve the client's well-being, the expert will not invest in improving the diagnostic precision, i.e. $E^{passive} = 0.5$ and $g(E^{passive}) = 0$.

3.3 Case 1: Active Client & Observable Precision

In this case, the client is active and participates in choosing treatments. The expert's diagnostic precision is observable to the client. Figure 2 describes the timing of the game for this case. Different from the benchmark case, after the expert makes the treatment recommendation in Step 4, the client in Step 5 observes the diagnostic precision E and the recommendation T^x , and then determines the actual treatment T^c to receive. Therefore, the client has the option to go against the expert by choosing the treatment opposite to the recommended one.

Lemma 1 *When $s^c = s^x$, the client always follows the expert's treatment recommendation regardless of the precision level E .*

Proof. Obvious. ■

Lemma 1 is intuitive: when the client and the expert hold consistent opinions about the problem, there is no reason for the client to overrule the expert's treatment recommendation. The client will consider overruling the expert's recommendation only when their private signal about the problem conflicts with the expert's diagnosis.

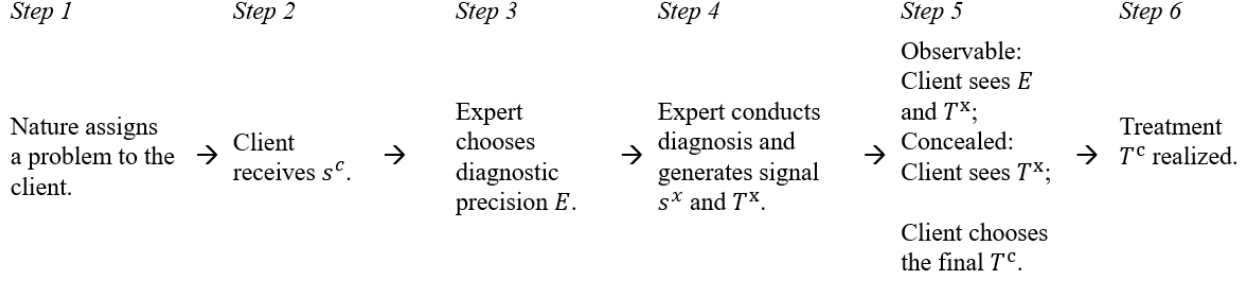


Figure 2: Game Procedure with Active Client: Observable and Concealed Cases

Lemma 2 *When $s^c \neq s^x$, the client follows the expert's treatment recommendation only when $E \geq q$. Otherwise, the client will revise the treatment into $T^c = s^c$.*

Proof. When $s^c \neq s^x$, the client's expected utility from following the expert is: $E^{follow}(U^p) = EH + (1 - E)L$. If not following the expert, the expected utility is: $E^{overrule}(U^p) = qH + (1 - q)L$. Since both E and q are between 0.5 and 1, and $H > L$, $E^{follow}(U^p) \geq E^{overrule}(U^p)$ only when $E \geq q$. ■

Lemma 2 describes the client's best strategy in response to the observed diagnostic precision E . It claims that when the expert's diagnostic result conflicts with the client's private signal, the client's compliance depends on the expert's diagnostic precision. If the expert's diagnosis is less precise than the client's privation signal, then the client will choose not to comply with the expert recommendation.

Proposition 2 *When the client is active and the expert's diagnostic precision is observable to the client: (1) For $\gamma < 4k(q - 0.5)/(H - L)$, the expert's equilibrium strategy is $E^{act\&obs} = 0.5$, and the client will only follow their private signal to choose treatment on this equilibrium path. (2) For $\gamma \geq 4k(q - 0.5)/(H - L)$, the expert's equilibrium strategy is $E^{act\&obs} = 0.5 + \gamma(H - L)/2k$, and the client will always follow the expert's treatment recommendation on this equilibrium path.*

Proof. To have the expert choose a diagnostic precision level greater than 0.5, two conditions must be satisfied: (i) The expert's expected utility is higher with this precision level than with $E = 0.5$; (2) The client is compliant with the expert. Notice that if the client is

fully compliant, then it goes back to the passive client case, so the candidate optimal choice for the expert under this case is $E^{act\&obs} = 0.5 + \gamma(H - L)/2k$. To satisfy (i), we need: $E[U^x(E^{act\&obs})] \geq E[U^x(0.5)]$, which can be simplified to $\gamma \geq 4k(q - 0.5)/(H - L)$. Now still need to show: $E^{act\&obs} > q$. Since $\gamma \geq 4k(q - 0.5)/(H - L)$, there will be: $E^{act\&obs} > 0.5 + [4k(q - 0.5)/(H - L)](H - L)/2k = 2q - 0.5$. Because $q \geq 0.5$, $E^{act\&obs} > q$ always hold with $\gamma \geq 4k(q - 0.5)/(H - L)$. ■

According to Proposition 2, the equilibrium depends on γ , which is the expert and client's incentive alignment, or in other words, the expert's motivation to solve the client's problem. If the expert's motivation is lower than the threshold (i.e., $\gamma < 4k(q - 0.5)/(H - L)$), the market reaches a "low-precision-low-compliance" equilibrium. If the expert has high enough concerns about the client, the expert will exert sufficient effort to increase the precision. The client observes the precision and fully complies. In this case, the market reaches a so-called "high-precision-high-compliance" equilibrium. This equilibrium outcome mirrors the benchmark case where the client is passive.

3.4 Case 2: Active Client & Concealed Precision

In this case, the client does not observe the expert's diagnostic precision, but still decides whether to comply with the recommendation. This mirrors real life more closely, as clients typically cannot gauge the effort an expert invests in the diagnosis or the precision of the diagnosis. However, after receiving the expert's recommendation, the client must decide whether to follow it. The procedure is identical to Case 1 with observable precision as shown in Figure 2, except that in Step 5, the client sees only the recommendation T^x , and chooses whether to comply.

Notice that in this case, Lemma 1 still holds: when the client's private signal is consistent with the expert's diagnosis, then the client's optimal choice is to follow the expert's treatment recommendation. However, in this case, since the client cannot observe the expert's diagnostic precision, when her private information conflicts with the expert's diagnosis, her

compliance will depend on the expectation of her belief, $\hat{E}$, about the precision level.

Lemma 3 *When $s^x \neq s^c$, the client follows the expert's treatment recommendation only when $\hat{E} \geq q$. Otherwise, the client will revise the treatment into $T^c = s^c$.*

Proof. Obviously similar to the proof of Lemma 2. ■

Lemma 3 is the client's best strategy when the diagnostic precision is not observable, and the private signal is inconsistent with the expert's diagnostic result. It predicts that the client's compliance depends on their belief $\hat{E}$: if the client believes their private signal is more accurate than the expert's diagnosis, they will follow their private signal when choosing the treatment. This is similar to a real-life scenario where a client does not comply with an expert's recommendation because they think the expert did not invest enough effort in the diagnosis (e.g., a doctor does not thoroughly examine all symptoms or analyze the test results). Instead, the client relies on their self-diagnosis to solve the problem. In this case, the expert's optimal choice is influenced not only by the client's private signal accuracy q , but also the relationship between q and $\hat{E}$:

Proposition 3 *When the client is active and the expert's diagnostic precision is not observable to the client: (1) For $\gamma < 4k(q - 0.5)/(H - L)$, the expert's equilibrium strategy is $E^{\text{act\&obs}} = 0.5$, and on this equilibrium path, the client only follows their private signal and chooses $T^c = s^c$. (2) For $\gamma \geq 4k(q - 0.5)/(H - L)$, there are two equilibria: (i) the client believes that $\hat{E} < q$, and on this equilibrium path, the expert chooses $E = 0.5$ while the client only follows their own private signal and chooses $T^c = s^c$; (ii) the client believes that $\hat{E} \geq q$, and on this equilibrium path, the expert chooses $E^{\text{act\&conceal}} = 0.5 + \gamma(H - L)/2k$ while the client always follows the expert's recommendation, i.e., chooses $T^c = T^x$.*

Proof. For $\gamma < 4k(q - 0.5)/(H - L)$, there will be $E[U^x(0.5 + \gamma(H - L)/2k)] < E[U^x(0.5)]$, so the expert will always choose $E = 0.5$, while the client's equilibrium strategy will be only following her private signal. For $\gamma \geq 4k(q - 0.5)/(H - L)$, if the client believes that $\hat{E} < q$, then they will not follow the expert, and the expert's equilibrium strategy is $E = 0.5$. Both

parties have no motivations to deviate. If the client believes that $\hat{E} \geq q$, then the expert’s equilibrium strategy is $E^{act\&conceal} = 0.5 + \gamma(H - L)/2k$ because the client is now “passive”, similar to the case in Proposition 1. ■

Proposition 3 is different from Proposition 2 in that when the diagnostic precision is concealed to the client, the model predicts two equilibria for an expert with alignment parameter $\gamma \geq 4k(q - 0.5)/(H - L)$: one where the expert invests in increasing diagnostic precision, and the client follows the expert’s recommendation, and another where the expert does not invest in improving diagnostic precision, and the client does not follow the expert’s recommendation. The first equilibrium returns identical outcomes as those predicted in Proposition 2 (the Active Client & Observable Precision case) as well as in Proposition 1 (the Passive Client case). The second equilibrium is unique to this scenario and does not exist in the other cases. This additional equilibrium suggests a scenario where, despite high incentive alignment, the expert and client end up with a low-precision-low-compliance outcome. Intuitively, this happens because highly motivated experts might perceive clients’ active participation as disregarding their advice, making their investment of effort a waste. In this equilibrium, compared to Case 1 and the benchmark case, clients in Case 2 receive fewer efforts from experts and are less likely to solve problems.

In summary, I demonstrate that when the client cannot assess the precision of the expert’s diagnosis (or, in other words, the expert’s effort in the diagnosis), two equilibria can coexist: one with high diagnostic precision and high compliance, and another with low diagnostic precision and low compliance. Compared to the case where clients are fully compliant (i.e., the passive client case), if the latter equilibrium occurs, giving clients the opportunity to overrule experts will make both parties worse off.

However, this theoretical framework does not provide criteria to predict the likelihood of the second equilibrium occurring. Therefore, I conduct a laboratory experiment to further investigate the existence of this equilibrium. In the next section, I will introduce the experimental design motivated by this theoretical framework, with specific parameter setups.

Based on the parameter setups, I will present the testable predictions.

4 Experimental Design

In this section, I first introduce the model parameters for the experiment. The choice of parameters is motivated by the research interest in verifying the impact of client’s active participation in treatment selection on the expert’s performance. Then I present the treatment conditions and further details of the experimental procedure. In the last part of this section, I discuss testable predictions based on the experimental parameters.

4.1 Parameters

For all experimental treatment conditions (details will be explained later), it is common knowledge that each client receives one of two problems (problem A or problem B) with equal probability. The client does not know the problem but receives a signal of either A or B, with the probability of 0.6 that this signal is consistent with her true problem ($q = 0.6$). The diagnostic result and treatment will also be either A or B. If the client receives the correct treatment, he/she will get 120 tokens ($H = 120$), and if not, he/she will get 20 tokens ($L = 20$)⁴.

The expert earns a lump sum of 80 tokens for each interaction with the client ($M = 80$). The purpose of this lump sum is to balance the expected payoff between experts and clients. The expert’s task is to choose the diagnostic precision $E \in \{0.50, 0.60, 0.70, 0.80, 0.90, 1.00\}$, incurring a cost given by $g(E) = 100(E - 0.5)^2$. To simplify the task, a cost table is presented to the expert as shown in Table 1.

⁴The exchange rate is 10 tokens = 1 dollar.

Table 1: The Expert’s Cost Table of Diagnostic Precision

Precision	50%	60%	70%	80%	90%	100%
Cost	0	1	4	9	16	25

Note: This is the cost table that subject experts observed when making the precision choice in each round.

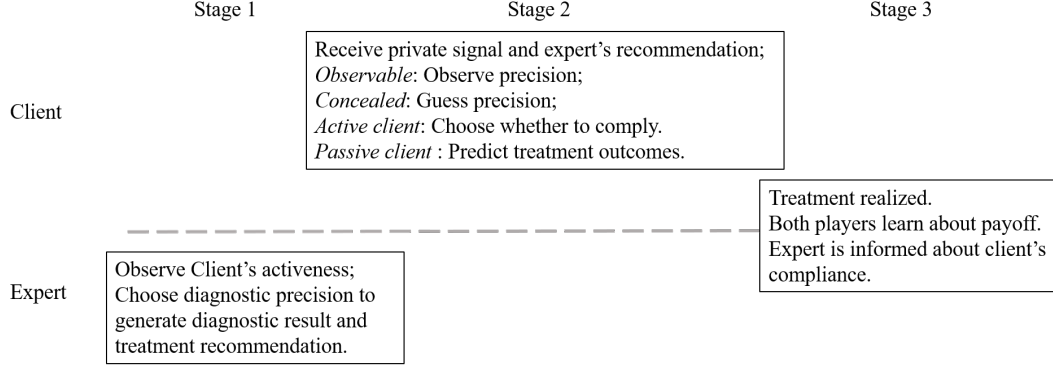


Figure 3: Game Procedure of a Round

4.2 Treatment Conditions

There are two treatment conditions: *Observable condition* and *Concealed condition*. In each condition, the expert-client interaction game repeats for 20 rounds, with random matching of experts and clients in each round.

First of all, to examine the role of clients' active participation in decision-making, I vary the opportunity for clients to choose treatments on their own. Within each session, clients are randomly assigned to be “active” with a probability of 70% and be “passive” with a probability of 30%, and their activeness status remains fixed for the whole session. Therefore in each session, experts interact with both types of clients. More subjects are assigned to be active because the active client case is a more important focus of this study.

To verify the propositions in the previous section, two treatment conditions, the *Observable condition* and the *Concealed condition*, vary the client's opportunity to observe diagnostic precision, at the between-subject level. Figure 3 shows the game procedures for these two treatment conditions, with details described below.

Observable Condition: In Stage 1, the expert observes the client's activeness and chooses

the (costly) precision level to generate a diagnostic result. The chosen precision level and the treatment recommendation from the diagnostic result are presented to the client in Stage 2, as well as the client’s private signal about the problem. If the client is active, then the client chooses whether or not to comply with the doctor. Therefore, the client can choose the treatment *opposite* to the expert’s recommended one if he/she likes. If the client is passive, the client does not do anything according to the theory. But in the experiment, I ask these passive clients to guess whether or not the problem will be solved, and a correct guess returns them 10 tokens. In Stage 3, the treatment is realized.

Concealed Condition: The game procedure of the Concealed condition differs from the Observable condition in that clients are not able to observe their matched experts’ diagnostic precision levels at Stage 2. Instead, I elicit the client’s belief about the precision level, which will help verify Proposition 3 that clients’ compliance largely depends on their own beliefs about diagnostic precision. A correct guess of the precision rewards the client 10 tokens.⁵

In addition to these two conditions, this experiment also contains two institutional changes, the *Communication condition* (clients and experts chat with each other) and the *Reputation condition* (clients share their ratings on experts), both pre-registered. The purpose of these two conditions is to explore the impact of giving clients (noisy) information about experts on both the clients’ problem-solving and experts’ investment in diagnostic precision. Details of the design and results are reported in Section B in the Appendix.

4.3 Experimental Procedure

Following previous practice (Fiedler et al., 2013), this experiment separates preference measures from the main laboratory experiment to reduce spillover effects. Before participating in the lab experiment, all subjects completed an online survey with two widely used, incentivized tasks to measure risk aversion and altruism.

The altruism measure adapts the dictator game suggested by Forsythe et al. (1994),

⁵Clients do not know whether their guesses are correct until the end of the whole experiment, to avoid the feedback of experts’ precision affecting clients’ subsequent beliefs and behaviors.

implemented with a “role uncertainty” matching approach: all subjects make resource allocation decisions, and are then randomly matched into pairs, with one member of each pair randomly chosen as the dictator whose allocation decision is implemented.⁶ To address the common issue of many participants choosing a 50/50 allocation, which can reduce variation in choices (as noted in Engel (2011) meta-analysis), I follow Andreoni and Miller (2002) by setting the relative price of giving to 0.5: for every 100 points available for allocation, each point the dictator keeps is worth 1 token to the dictator, but each point allocated to the recipient is worth 2 tokens to the recipient. Screenshots of the instruction and decision scene (Figures E2 and E3) are in Appendix E. This revision not only reduces the mass concentration of 50/50 allocations, but also makes the dictator’s decisions similar to the expert’s decisions – in both cases, the benefit received by the recipient or client from the dictator’s giving or the expert’s effort is greater than the cost to the dictator or expert.

The risk preference is measured by the investment task suggested by Charness and Gneezy (2010). Subjects choose how much of a fixed endowment to invest in a risky asset, which gives a continuous measure of risk preferences. This measure is desirable due to its simplicity and ease of use with various populations (Charness et al., 2013).

Each lab session is assigned to one of the four treatment conditions discussed earlier. Subjects are randomly assigned to the role of either the expert or the client. Within each role, clients are randomly assigned to be “passive” or “active”, and experts are randomly assigned with the alignment parameter γ . Those role assignments remain fixed throughout the experiment. The matching of experts and clients is random in each round to prevent the construction of reputation through repeated interaction. To further reduce any interdependency between rounds while maintaining incentive compatibility, subjects are informed at the start that one out of 20 rounds will be randomly selected for payment (see Charness et al. (2013) and Azrieli et al. (2018) for discussion on the incentive compatibility of the random

⁶Iriberry and Rey-Biel (2011) pointed out that the use of role uncertainty may overestimate the prevalence of social-welfare-maximizing preferences within the sample. However, since I am interested in the correlation between altruism and experts’ choices in diagnostic precision rather than the overall level of other-regarding preferences, this method provides credible measures in my context.

payment scheme). At the end of the experiment, subjects complete a questionnaire collecting their demographic information and their feedback about the experiment. This study is pre-registered at AsPredicted.org, with the pre-registration file attached in Appendix C.⁷

4.4 Predictions

This subsection presents predictions under the Observable and Concealed conditions, which are testable by a laboratory experiment.

4.4.1 Prediction of Diagnostic Precision with Passive Clients

Figure 4(a) illustrates Prediction 1, where the dashed line is the prediction of the expert’s optimal choice of diagnostic precision when interacting with a passive client: $E^* = 0.5 + \frac{\gamma}{2}$. This figure indicates that the optimal diagnostic precision is increasing with the incentive alignment parameter γ . This prediction is a numeric case of Proposition 1 derived from the parameter setup of the experiment. Note that this prediction holds for both the Observable and Concealed conditions, as the client’s passivity means that their ability to observe the expert’s diagnostic precision does not influence the expert’s choice.

Prediction 1 *When interacting with passive clients, experts’ diagnostic precision is increasing with γ , from $E = 0.5$ to $E = 1$.*

4.4.2 Predictions of Diagnostic Precision with Active Clients

To predict expert’s investment in diagnostic precision for active clients, we should first check active clients’ strategies. Consistent with the prediction from Lemma 1, under both Observable and Concealed conditions, clients are predicted to always follow their experts when

⁷Please note that while the pre-registration was framed using a patient-doctor context, the actual laboratory experiment was conducted using a neutral expert-client context, as detailed in the game instructions in Appendix D. This adjustment was made to enhance generalizability, ensuring that our findings apply beyond the medical context to broader expert-client interactions. Additionally, the pre-registration included an examination of the client’s willingness to pay to become an active client. However, after piloting a few sessions, we decided to remove this aspect to concentrate this study on the influence of client activeness on experts, avoiding the introduction of additional elements that may confuse subjects.

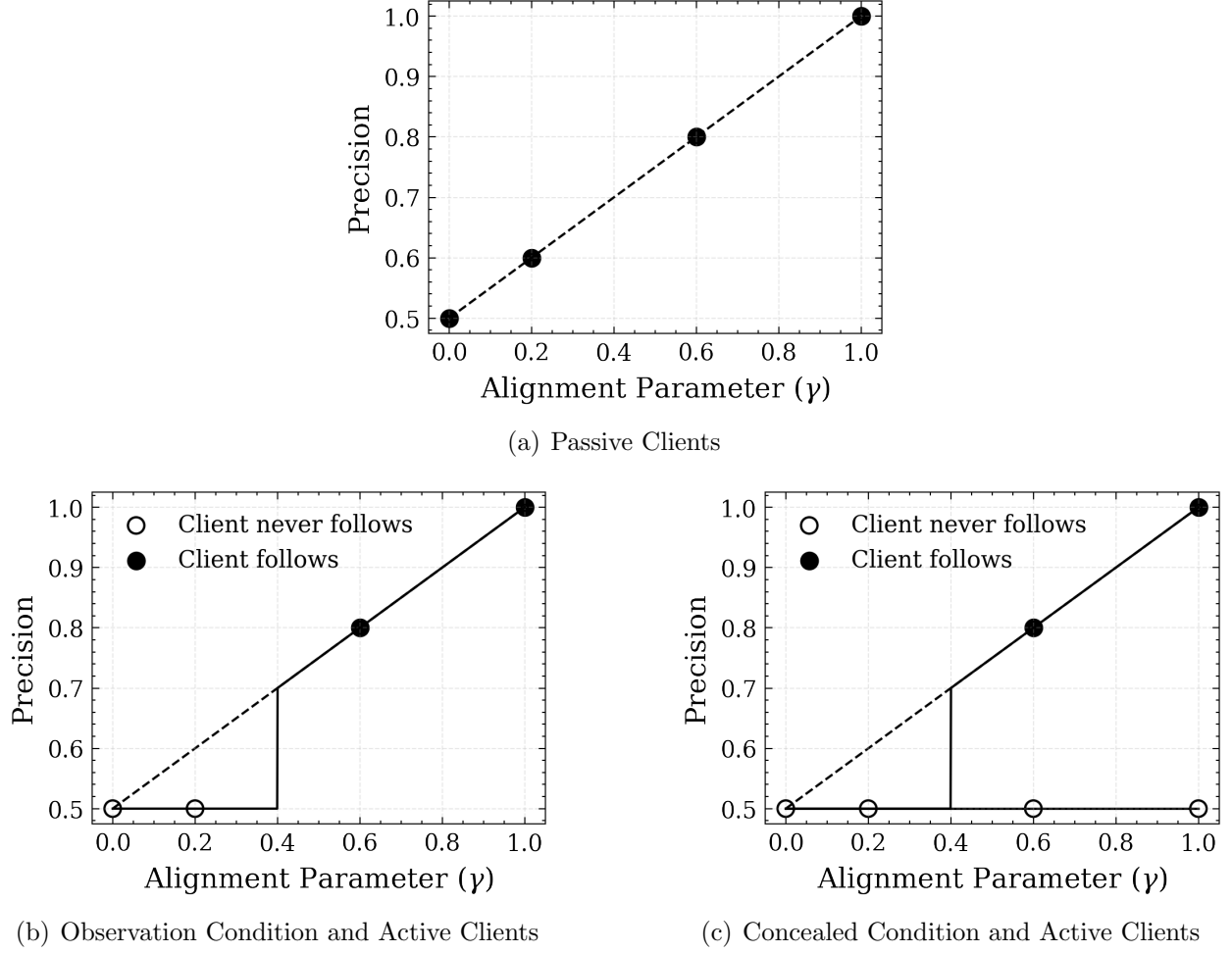


Figure 4: Equilibrium Predictions

Note. Dots in three panels highlight the predicted precision level choices for the four alignment parameters $\gamma \in \{0, 0.2, 0.6, 1\}$ used in the lab. In Panels (b) and (c), hallow dots indicating the equilibrium where clients believe that $\hat{E} < 0.6$, and solid dots indicating the equilibrium where clients believe that $\hat{E} \geq 0.6$. The dashed lines in Panels (b) and (c) depict the prediction of experts' diagnostic precision when clients are passive (i.e. from Panel (a)), as a benchmark for comparison.

their private signals are consistent with experts' diagnostic results. When an information conflict occurs, following Lemmas 2 and 3, under the Observable/Concealed condition, an active client is predicted to go against the expert if she observes/believes that the diagnostic precision is lower than 0.6, i.e. the precision of the client's private signal. In the rest of this subsection, I will focus on the case when there exists an information conflict between a client's private signal and an expert's diagnostic result.

For convenience of discussion, I name experts with $\gamma = 0$ or 0.2 as "low-alignment ex-

perts”, and experts with $\gamma = 0.6$ or 1 as “high-alignment experts”, indicating different levels of concerns that these experts have about clients’ well-being. This is because the threshold $\gamma = 4k(q - 0.5)/(H - L)$ in Propositions 2 and 3 equals 0.4 based on the numerical setup of the experiment, and experts with $\gamma < 0.4$ and experts with $\gamma \geq 0.4$ are predicted to behave differently, which will be explained below.

Figure 4(b) depicts the equilibrium prediction of the expert’s diagnostic precision and the client’s compliance under the Observable condition. Low-alignment experts ($\gamma = 0$ or 0.2) are predicted to choose the lowest diagnostic precision ($E = 0.5$), making no investment in improving precision, and their matched clients never follow their recommendations.⁸ In contrast, high-alignment experts ($\gamma = 0.6$ or 1) are predicted to invest to improve the diagnostic precision, following $E^* = 0.5 + \frac{\gamma}{2}$. For these experts, the outcomes are identical to those in the passive client case, meaning their diagnostic precision is predicted to be the same for both active and passive clients under the Observable condition.

Figure 4(c) characterizes the equilibrium prediction under the Concealed condition, derived from Proposition 3. Same as the Observable condition, for low-alignment experts under the Concealed condition, there exists only one equilibrium where clients believe that $\hat{E} < 0.6$. On this equilibrium path, experts choose the lowest diagnostic precision, and clients never follow their treatment recommendations. However, for high-alignment experts, two equilibria coexist under the Concealed condition. In the first equilibrium, clients believe that $\hat{E} \geq 0.6$ and always follow experts, while experts choose diagnostic precision $E^* = 0.5 + \frac{\gamma}{2}$. This equilibrium returns the outcome identical to the Observable condition and the passive client case. In the second equilibrium, clients believe that $\hat{E} < 0.6$, where experts choose the lowest precision and clients never follow experts. Therefore, under the Concealed condition, there exists an additional “low-compliance-low-precision” equilibrium where high-alignment

⁸One may notice that experts with $\gamma = 0.2$ are predicted to choose the lowest precision of 0.5 , which differs from the passive client case where these experts still invest a positive amount to increase the diagnostic precision. Intuitively, this is experts face a tradeoff when interacting with active clients: they can either invest to improve the diagnostic precision and make clients better off, or choose not to exert any effort and have clients rely on their private signals to solve problems. Given low-alignment experts’ low concerns about clients’ welfare ($\gamma < 4k(q - 0.5)/(H - L)$), they choose the latter strategy.

experts choose the lowest diagnostic precision level.

The existence of this additional low-compliance-low-precision equilibrium indicates that high-alignment experts may treat passive and active clients differently under the Concealed condition. This is testable by an experiment: if this additional equilibrium exists, then the average diagnostic precision received by active clients will be lower than the passive clients, conditional on those pairs with high-alignment experts; furthermore, as experts' incentive alignment increases from 0.6 to 1, their investment gap between active and passive clients also increases. Importantly, as previously discussed, this investment gap between active and passive clients is not expected to exist under the Observable condition. These predictions are summarized in Prediction 2.

Prediction 2 *(1) Conditional on Observable condition: high-alignment experts ($\gamma = 0.6$ or 1) choose the same diagnostic precision level for active and passive clients; (2) Conditional on the Concealed condition: high-alignment experts ($\gamma = 0.6$ or 1) choose a lower diagnostic precision level for active clients than for passive clients.*

Intuitively, the prediction above implies that experts who are more incentive aligned with clients may be more discouraged by clients' active participation in diagnosis and problem solving. For example, doctors with a strong motivation to solve patients' problems may be particularly concerned about patient compliance, especially when their expertise is questioned; similarly, mentors who care about mentees may hesitate to invest further effort when their advice is not taken seriously. The theory further implies that this adverse effect disappears if experts can credibly demonstrate their diagnostic precision. However, in many real-world expert-service settings, diagnostic precision is not directly observable. Experts cannot easily prove the accuracy of their diagnosis, and thus may feel discouraged by clients' noncompliance. This makes the adverse effect of active client participation a robust concern in practice.

Finally, Predictions 3 and 4 are secondary predictions for the behaviors of active clients.

Prediction 3 *Clients follow their experts if their private signal is consistent with experts’ diagnostic precision.*

Prediction 4 *If there exists a conflict between the client’s private signal and the expert’s diagnostic result, clients overrule experts if they observe (believe) that the experts’ diagnostic precision is higher than 0.6 under the Observable (Concealed) condition.*

5 Results

A total of 436 undergraduate students participated in the experiment, with their basic background information reported in Table 2. They were randomly assigned to one of the four treatment conditions discussed in the previous section. The average earning from lab sessions is \$20.03, including \$10 participation fee. In addition, subjects on average received \$1.90 for the online survey prior to lab sessions.

Table 2: Summary Statistics of Subjects’ Background Information

	Treatments				
	All	Observable	Concealed	Communication	Reputation
N. of Subjects	436	98	130	104	104
% Female	58.03	54.08	63.85	54.81	57.69
Age	20.11	19.71	19.91	20.41	20.42
	(1.85)	(1.48)	(1.47)	(1.83)	(2.44)
Risk Measure	49.80	53.95	52.13	46.04	46.74
	(26.70)	(27.11)	(28.47)	(25.06)	(25.10)
Altruism Measure	4.22	4.15	3.95	4.63	4.22
	(2.36)	(2.33)	(2.43)	(2.38)	(2.26)

Note: Standard deviations in parentheses. Risk measure is the risk tolerance measured by the investment decision in [Charness and Gneezy \(2010\)](#), ranging from 0 to 100, with higher values indicating higher risk tolerance. Altruism measure is from the dictator game suggested by [Forsythe et al. \(1994\)](#), with values ranging from 1 to 11 and higher values indicating a higher altruism level.

5.1 Expert Diagnostic Precision in Observable and Concealed Conditions

The primary outcome of interest is experts' *diagnostic precision*, which reflects their investment of effort in achieving a certain level of diagnostic precision. Table 3 displays the average diagnostic precision conditional on treatment conditions and experts' alignment parameter γ . Panel A is restricted to those rounds with active clients, and Panel B focuses on passive clients. This table also reports the p -values from linear regressions comparing treatment conditions, clustered at an individual level.

The key pattern from Table 3 is that for all treatment conditions, the average diagnostic precision increases with the alignment parameter γ . This pattern is consistent with Prediction 1, as well as Figures 4(b) and 4(c), indicating that experts with higher incentive alignment with clients choose higher diagnostic precision.

The statistical tests between Observable and Concealed conditions in Table 3 show no differences in the diagnostic precisions between the Observable and Concealed conditions, conditional on active/passive clients. This finding seems to not offer support for the existence of the additional low-compliance-low-precision equilibrium in the Concealed condition as characterized in Figure 4(c). However, Prediction 2 offers an approach to verify the existence of the low-compliance equilibrium by comparing the active and passive clients within the Concealed condition: the low-compliance equilibrium among active clients will make high-alignment experts choose a lower average diagnostic precision for active clients than for passive clients. This comparison is also crucial for investigating the influence of client's activeness on expert's investment in the diagnostic precision.

To directly compare diagnostic precision for passive versus active clients, Figure 5 depicts the average diagnostic precision by treatment conditions, client activeness, and experts' alignment parameter γ . In Figure 5(a), under the Observable condition, experts with $\gamma = 0.6$ or 1 show no statistically significant difference in the average diagnostic precision for passive

Table 3: Expert Average Diagnostic Precision

Panel A: Diagnostic Precision for Active Clients

	$\gamma = 0$	$\gamma = 0.2$	$\gamma = 0.6$	$\gamma = 1$	All
Observable	0.531 (0.081)	0.718 (0.139)	0.805 (0.138)	0.828 (0.181)	0.710 (0.179)
Concealed	0.599 (0.164)	0.642 (0.156)	0.756 (0.153)	0.771 (0.176)	0.696 (0.176)
<i>p</i> -value: Concealed vs. Observable	0.147	0.143	0.255	0.460	0.669

Panel B: Diagnostic Precision for Passive Clients

	$\gamma = 0$	$\gamma = 0.2$	$\gamma = 0.6$	$\gamma = 1$	All
Observable	0.573 (0.141)	0.758 (0.130)	0.853 (0.125)	0.828 (0.175)	0.740 (0.180)
Concealed	0.607 (0.198)	0.730 (0.146)	0.875 (0.142)	0.894 (0.150)	0.787 (0.192)
<i>p</i> -value: Concealed vs. Observable	0.619	0.563	0.594	0.401	0.188

Note: Standard deviations are in parentheses. *p*-values are from linear regressions using diagnostic precision as the outcome variable and the indicator of the two treatment conditions for comparison as the independent variable, clustered at the individual level. For each regression, I report the *p*-value associated with the estimated coefficient for the independent variable. These *p*-values indicate the statistical significance for the null hypothesis that the diagnostic precision is identical between the two conditions for comparison, accounting for repeated observations from each subject.

vs. active clients. This finding aligns with Prediction 2, indicating that when the diagnostic precision is observable, high-alignment experts treat active and passive clients equally in terms of diagnostic precision. In contrast, in the Concealed condition depicted in Figure 5(b), the average diagnostic precision from high-alignment experts for active clients is statistically significantly lower than that for passive clients. This finding further supports Prediction 2 that when the diagnostic precision is not observable, high-alignment experts treat passive and active clients differently, with active clients receiving lower diagnostic precision.

To further confirm findings from Figure 5, Table 4 presents results from regressions with diagnostic precision as the dependent variable. Model (1) includes independent variables of the expert’s alignment parameter, the matched client’s activeness, the treatment conditions, and the interactions of these variables. Model (2) further controls for rounds and individual characteristics including gender, ethnicity, altruism measure, risk attitude measure, whether the subject is of economics major, and whether the subject comes from Texas.

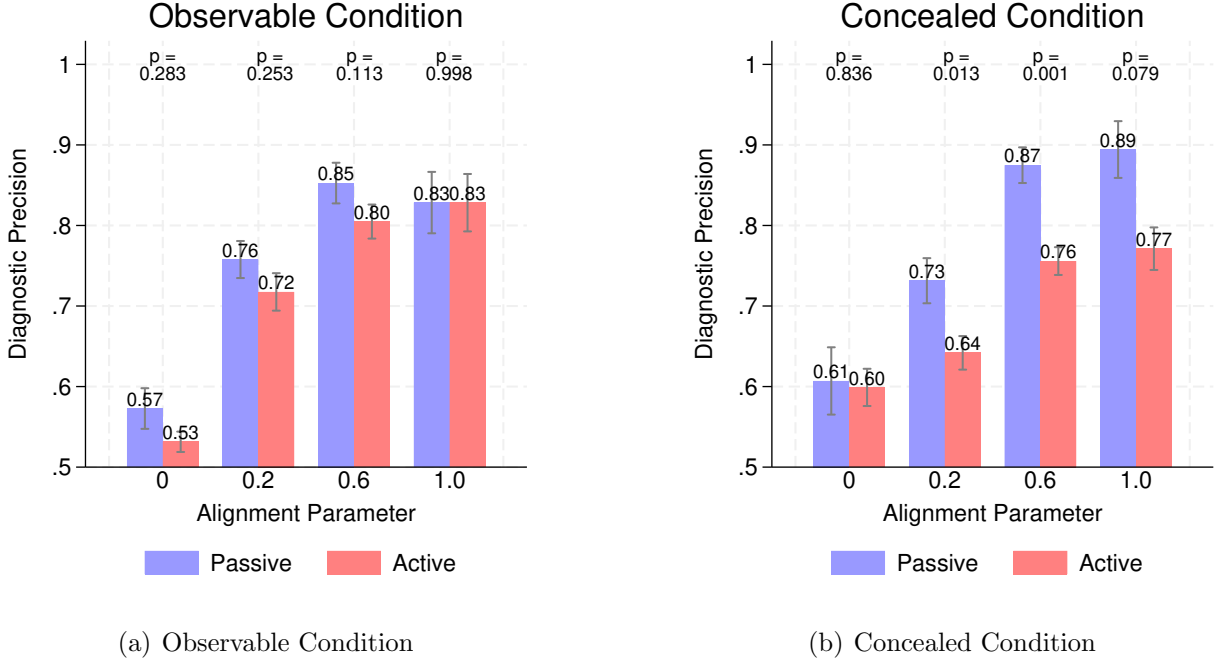


Figure 5: Experts' Average Diagnostic Precision: Active vs. Passive Clients

Note: Error bars are the 95% confidence intervals for averages. p -values are from linear regressions using diagnostic precision as the outcome variable and the binary indicator that equals 1 for active clients and 0 for passive clients as the independent variable, clustered at individual level. For each regression, I report the p -value associated with the estimated coefficient for the active-client indicator. These p -values indicate the statistical significance for the null hypothesis that the diagnostic precision is identical between the active and passive clients, accounting for repeated observations for each subject.

The regression analysis yields several findings. First, the positive and statistically significant coefficient for the alignment parameter indicates that experts' diagnostic precision increases with their incentive alignment with clients, consistent with Predictions 1. Second, the estimated coefficient for the active client indicator shows that clients' active participation significantly reduces diagnostic precision by about 5 percentage points. These findings remain robust when additional controls are included in Model (2).

Due to the complexity of interpreting multiple interaction terms in Table 4, based on Model (2), I perform a marginal effect analysis of the "Active" indicator to investigate the effect of client activeness on diagnostic precision conditional on each treatment condition and each alignment parameter level. Figure 6 visualizes the analysis results. This figure shows that for low-alignment experts ($\gamma = 0$ or 0.2), the negative impact of the client's

Table 4: Linear Regression on Experts' Diagnostic Precision

	DV: Diagnostic Precision	
	(1)	(2)
Alignment Parameter	0.247*** (0.064)	0.206*** (0.061)
Concealed	0.010 (0.049)	0.037 (0.040)
Concealed $\times$ Alignment Parameter	0.054 (0.095)	0.054 (0.087)
Active	-0.051* (0.026)	-0.054** (0.021)
Active $\times$ Alignment Parameter	0.045 (0.046)	0.053 (0.043)
Concealed $\times$ Active	0.009 (0.040)	0.005 (0.033)
Concealed $\times$ Active $\times$ Alignment Parameter	-0.155* (0.084)	-0.140* (0.074)
Constant	0.644*** (0.033)	0.653*** (0.086)
Individual Controls	No	Yes
Round	No	Yes
Observations	2280	2280
Number of Individuals	114	114

Note: Individual controls include gender, ethnicity, whether the subject comes from Texas, subjects' altruism measured by choices in the dictator game, subjects' risk tolerance measured by choices in the investment game, and indicator of whether subjects major in economics or agricultural economics.

Standard error in parentheses, clustered at an individual level. * $p < .1$; ** $p < .05$;

*** $p < .01$.

activeness on the diagnostic precision is similar under the Concealed and Observable conditions. For high-alignment experts ($\gamma = 0.6$ or 1), the negative impact from the client's activeness is larger under the Concealed condition, but is nearly eliminated to zero under the Observable condition. This finding aligns with the findings from Figure 5 that under the Concealed condition, the gap in investment in active and passive clients increases with experts' incentive alignment. This finding also provides evidence of the existence of the additional low-compliance-low-precision equilibrium where high-alignment experts choose the lowest diagnostic precision, as discussed in Subsection 4.4.2.

In the realm of healthcare, this finding suggests that in an ideal scenario where a patient

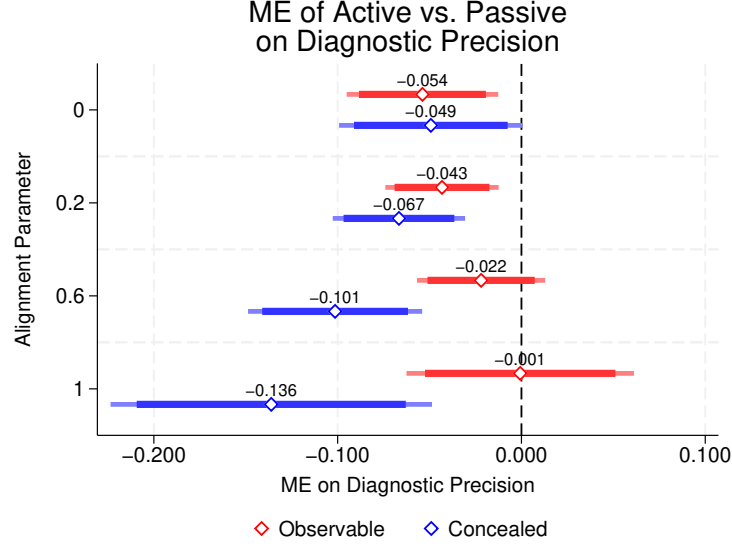


Figure 6: Marginal Effect of Client Activeness on Diagnostic Precision

Note: This figure is the marginal effect analysis derived from Model (2) in Table 4. The width of bars are 95% and 90% confidence intervals.

can observe how much effort a doctor exerts to achieve a certain diagnostic precision, an altruistic doctor will not worry about patients not following their advice. This is because the doctor knows that the patient, after observing the doctor's effort put into achieving a highly precise diagnosis, will be persuaded to follow the doctor's advice. Conversely, in a more realistic setting where patients are unable to assess the diagnostic precision of a doctor, doctors with stronger concerns about patients' well-being may worry that their advice will not be taken seriously. Patients' active participation in treatment selection can lead to concerns that they may overrule doctors' recommendations, making doctors' diagnostic efforts seem wasted. Consequently, doctors who care deeply about patients' health are more sensitive to whether a patient is fully compliant. Compared to patients who always follow their advice, doctors tend to invest less effort in diagnosing patients they believe will not fully adhere to their recommendations.

5.2 Client Active Participation and Problem Solving

Given the difference in diagnostic precision that experts choose for active versus passive clients, the next question is whether active participation actually reduces a client’s chance of solving the problem. Before examining this outcome, it is important to first investigate whether and how clients participate in treatment selection in the experiment.

Figure 7(a) shows the proportion of compliance among active clients by whether there exists an information conflict between the client’s private signal and the expert’s diagnostic result, pooling the Observable and Concealed conditions together. Consistent with Prediction 3, when there is no information conflict, active clients comply with their experts 91% of the time. When an information conflict occurs, this proportion reduces to 64%. Additionally, Figure 7(b) restricts the data to those rounds with information conflicts and shows that there is no statistically significant difference in the proportion of compliance between the Observable and Concealed conditions.

Recall that Prediction 4 states that when an information conflict occurs, active clients will overrule experts if the observed/believed diagnostic precision is lower than their private signal precision (0.6). Figure 7(c) shows that under the Observable condition, when clients observe a diagnostic precision of 0.5, only 14% of clients comply with their matched experts, and this compliance proportion increases to 97% when the diagnostic precision rises to 1. Similarly, Figure 7(d) presents comparable information for the Concealed condition, where clients’ compliance increases with their beliefs about the diagnostic precision.⁹

Overall, the four panels in Figure 7 indicate that active clients’ compliance decisions are consistent with the testable predictions. In other words, subjects who were assigned the role

⁹One may notice that when the belief about precision $\hat{E} = 1$, only 76% of subjects comply with the expert, while they ought to fully follow the expert if they believe that the expert is 100% precise. I noticed from the data that there are several subjects who did not make consistent choices between belief elicitation and compliance: they frequently believed in $\hat{E} = 1$ but chose to overrule the expert. For example, there is a subject with label 3009 who always chose $\hat{E} = 1$ but never followed. In Appendix Figure A1 I present again the figure by excluding this subject 3009. However, this does not mean that the result is not robust or a single subject can change the result. In the Concealed condition, subjects believe that $\hat{E} = 1$ for only 13.6% of time, which makes the data sensitive to a single subject’s frequent choices of $\hat{E} = 1$.

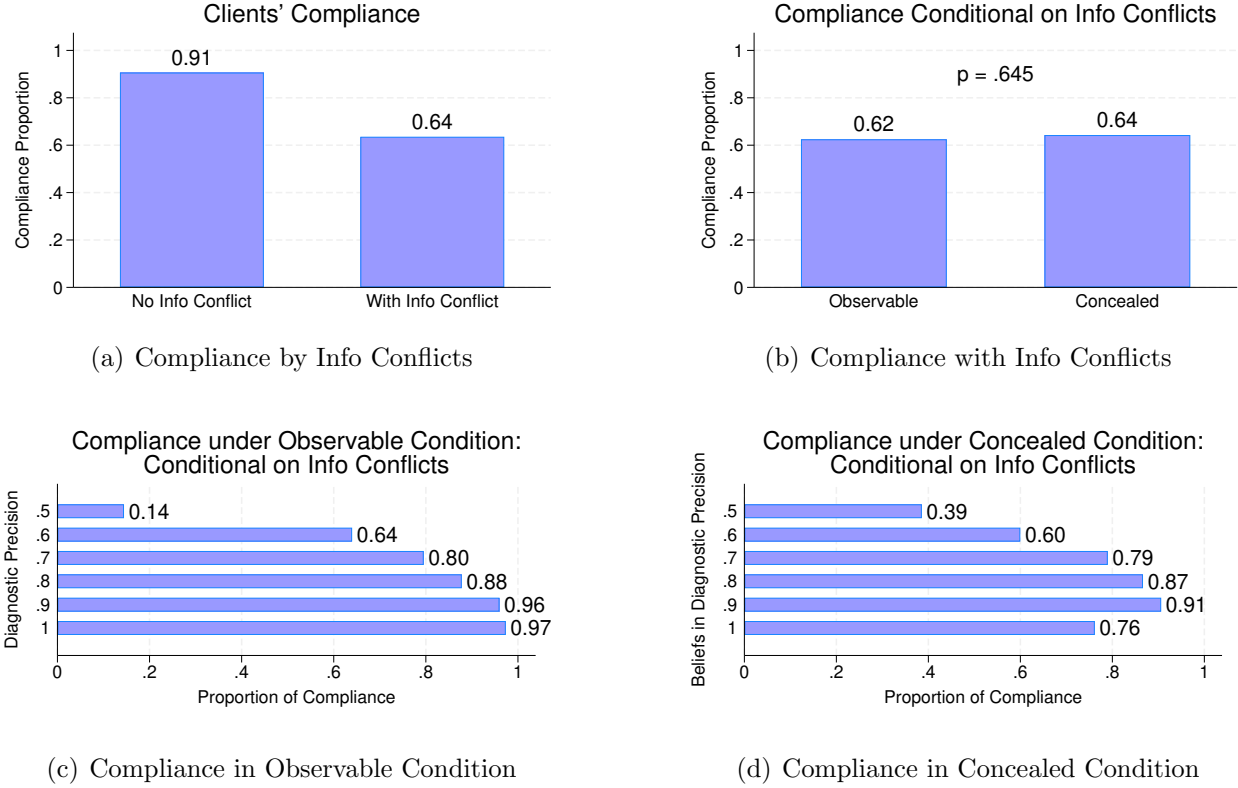


Figure 7: Information Conflicts and Client Compliance

Note: Data is restricted to Active clients in Observable and Concealed conditions. “No Info Conflict” means the client’s private signal is consistent with the expert’s diagnostic result. “With Info Conflict” means the client’s private signal differs from expert’s the diagnosis. “Compliance” means that the active client chooses the treatment that the expert recommends.

of client in the lab go against their experts for valid reasons. Next, I examine the influence of client activeness on their problem-solving.

Table 5 reports the proportion of solved problems for each treatment condition, separated by active and passive clients. This table conveys two key points. First, the impact of client activeness on problem-solving depends on the observability of diagnostic precision. Specifically, under the Observable condition, there is no statistically significant difference in the proportion of solved problems between active and passive clients. However, in the Concealed condition, active clients solve significantly fewer problems than passive clients. Second, active clients in the Concealed condition solve fewer problems compared to those in the Observable condition, while passive clients solve more problems in the Concealed

Table 5: Proportion of Solved Problems for Clients

	Proportion of Solved Problems (%)		<i>p</i> -value: Active vs. Passive
	Active Clients	Passive Clients	
Observable Condition	69.11	72.62	0.141
Concealed Condition	61.48	82.14	< 0.001
<i>p</i> -value: Concealed vs. Observable	0.013	0.013	

Note: *p*-values for activeness comparison (the last column): linear regressions are performed using the indicator of the problem solved as the dependent variable and the indicator of the client being active as the independent variable, clustered at an individual level, with the *p*-values of the coefficient estimate reported. *p*-values for treatment condition comparison: linear regressions are performed using the indicator of the problem solved as the dependent variable and the indicator of treatment conditions as the independent variable, clustered at an individual level, with the *p*-values of the coefficient estimate reported.

condition than in the Observable condition. These findings suggest that clients’ activeness is crucial in predicting their likelihood of solving problems. (Discussion of the Communication and Reputation conditions in this table will be saved to Subsection [B.3.1.](#))

To further verify the findings above, I perform a logit regression using a binary indicator of a problem being solved as the dependent variable. Independent variables of the regression include the binary indicator of client’s activeness (vs. passive), the treatment conditions (=1 if Concealed; =0 if Observable), and the interactions between these two variables. The marginal effect of client activeness from logit regressions are reported in Table [6](#). For the raw logit regression result, see Table [A1](#) in the appendix. This table suggests that conditional on the Observable condition, being an active client does not reduce the probability of solving a problem. However, under the Concealed condition, the active role reduces clients’ probability of solving their problems by 20.7 percentage points.

There are two reasons why active clients are less likely to solve problems in the Concealed condition. Firstly, as discussed earlier, high-alignment experts under the Concealed condition choose lower diagnostic precision for active clients compared to passive clients. Secondly, active clients in the Concealed condition are not able to accurately predict experts’ diagnostic precision (see Figure [A2](#) in the Appendix, which is a joint distribution of belief and actual

Table 6: Marginal Effect of Clients' Activeness on Problem Solving

	DV: Indicator of Problem Solved	
	(1)	(2)
Active vs. Passive (Treatment = Observable)	-0.035 (0.034)	-0.056 (0.034)
Active vs. Passive (Treatment = Concealed)	-0.207*** (0.032)	-0.207*** (0.032)
Individual Controls	No	Yes
Round	No	Yes
Observations	2280	2280
Number of Individuals	114	114

Note: This table shows the marginal effect of clients' activeness on the probability of solving a problem. The marginal effects are derived from logit regressions using the binary indicator of the problem solved (vs. not solved) as the dependent variable, with standard error clustered at individual level. Independent variables of the regression include the binary indicator of client's activeness (vs. passive), the treatment conditions (=1 if Concealed; =0 if Observable), and the interactions between these two variables. Individual controls include gender, ethnicity, whether the subject comes from Texas, subjects' altruism measured by choices in the dictator game, subjects' risk tolerance measured by choices in the investment game, indicator of whether subjects major in economics or agricultural economics.

Standard errors are in paratheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

diagnostic precision, reflecting that clients' beliefs are imprecise). Their imprecise beliefs lead to a higher probability of making mistakes. Table 7 provides further evidence supporting this argument. In this table, "Mistaken Compliance" refers to a situation where a client follows an expert but fails to solve the problem, indicating that the client should have overruled the expert. Conversely, "Mistaken Noncompliance" occurs when a client overrules an expert but should have followed their advice. Under the Concealed condition, 29.99% of the rounds with compliance are Mistaken Compliance, and 71.04% of the rounds with noncompliance are Mistaken Noncompliance. Both proportions are significantly higher than under the Observable condition (24.71% and 51.97% respectively). These differences highlight the higher rate of decision failures among active clients under the Concealed conditions than under the Observable condition. As a result, when information conflicts occur, the proportion of solved problems is lower in the Concealed condition (49.49%) than in the Observable condition (69.11%).

Given these findings, it becomes evident that the lack of transparency in the Concealed

Table 7: Active Clients’ Compliance and Welfare

	Observable	Concealed	<i>p</i> -value
% Mistaken Compliance	24.71	29.99	0.055
% Mistaken Noncompliance	51.97	71.04	0.001
% Solved Problems (Without Conflicts)	70.72	71.11	0.908
% Solved Problems (With Conflicts)	69.11	49.49	<.001

Note: *p*-values are from *t*-tests between the Observable and Concealed conditions; “% False Compliance” indicates the proportion of unsolved problems among those rounds where clients follow experts’ advice; “% False Noncompliance” indicates the proportion of unsolved problems among those rounds where clients overrule experts’ advice; “With Conflicts” indicates those rounds where clients’ private signals are not consistent with experts’ diagnostic results, while “Without Conflict” are those rounds without such information conflicts.

condition significantly hampers problem-solving effectiveness for active clients. In the Appendix B, I discuss the results of two additional treatments, the Communication and Reputation conditions, demonstrating how active clients can improve their problem-solving by gathering more information about experts, either directly through conversation, or through a shared rating system.

6 Discussions & Conclusion

This paper offers a comprehensive theoretical framework to discuss the influence of clients’ participation in decision-making on expert diagnosis. I conducted a lab experiment to verify the predictions derived from the theoretical framework. The experimental result provides supporting evidence of the adverse effect of clients’ active participation on expert effort and client well-being.

The most important finding from this experiment is that when clients cannot evaluate the effort experts exert to improve diagnostic precision, their active participation in treatment selection discourages experts from putting in more effort for a precise diagnosis. This discouraging effect is more pronounced among experts with higher incentive alignment with clients, i.e. with a stronger motivation to solve problems for clients. Intuitively, this occurs because experts view client participation as potential noncompliance with their advice, and

those more aligned with client interests are particularly sensitive to such noncompliance. Consequently, clients are less likely to solve their problems when they actively participate in treatment selection. To my best knowledge, the findings above have not been documented by empirical studies before, probably due to the challenges in observing expert effort and client participation in survey or administrative data.

In the realm of healthcare, one can interpret the expert-client incentive alignment as doctors' altruistic motivation to cure patient illness. Findings from this study suggest that in an ideal scenario where doctors can clearly demonstrate their diagnostic efforts and diagnostic precision, patients' self-diagnosis and participation in medical treatment selection do not discourage doctors from investing more effort. However, in a more realistic setting where patients cannot assess diagnostic precision, doctors with stronger motivation to cure patient illness have stronger concerns about their advice not being taken seriously. While these doctors still invest more effort in diagnosis compared to less altruistic ones, they show greater variance in their effort based on patient compliance. Thus, patients who make their own decisions receive less effort from doctors compared to fully compliant patients. However, one should notice that these findings do not suggest patients not participate in medical decision-making. Rather, I demonstrate the complex dynamics of expert-client interactions and point out a potential downside of client active involvement.

The findings can also be applied to a broader range of expert service industries. We can consider the alignment parameter in this study as a spectrum of expert-service contracts, varying in how closely expert incentives align with client incentives. In industries like automobile repair, experts' incentives often have limited dependence on clients' long-term satisfaction, as mechanics' compensation is secured upon service delivery. My research findings suggest that in such settings, experts are less concerned with clients' active participation once payment is guaranteed. Conversely, in fields like fiduciary services, where solving clients' problems directly impacts experts' welfare, fiduciaries are both legally and ethically obligated to prioritize clients' interests. Here, incentives for providing high-quality advice

align with client well-being, and experts might view client involvement as undermining their professional authority.

In this study, I also investigate the effectiveness of two institutional changes, Communication and Reputation, in improving clients' problem-solving. Both conditions reduce experts' investment gap between active and passive clients and demonstrate significant improvements in active clients' outcomes, albeit through different mechanisms. Under the Communication condition, direct conversation with experts leads to more accurate client beliefs about diagnostic precision, guiding clients to follow precise recommendations and disregard imprecise ones. The Reputation condition relies on a rating system, which provides valuable guidance through client ratings. Clients use these ratings to make better decisions about compliance, ultimately improving problem-solving.

As discussed in the literature review section, rating systems for medical services are under-used compared to those for movies, restaurants, and books. This study suggests that a reliable rating system could be valuable in medical services. While patients may lack specialized skills in medical diagnosis, their aggregated ratings can still reflect useful information about doctors' service quality. Compared to training programs aimed at improving patient-doctor communication, a reliable online rating system may potentially be a less costly approach to enhance patient well-being.

One limitation of the current study is that for the purpose of clarity and to isolate the impact of client active participation, we present a simplified model that only discusses two unknown problems with two corresponding solutions. While we believe that these limitations do not affect the primary findings and contributions of this study, future work can explore the model's robustness when incorporating factors like the distribution of unknown problems and the relationship between problems and solutions. For example, in some real-world scenarios, problems range from mild to severe, and more aggressive solutions can solve both mild and severe issues. This setup, which has been widely discussed in the credence goods literature, would enable richer discussions on the influence of client participation on expert diagnosis

and advice strategies.

Another limitation of that this study was conducted in a controlled laboratory environment using neutral language. While this method provides clear causal inference for the key variables, verifying the findings in the field would be an important extension. For example, previous studies have sent fictitious patients to clinics, showing that patient features like socio-economic status and patient demand can affect doctors' diagnoses and prescriptions (Currie et al., 2014; Gottschalk et al., 2020). Such practices will allow us to account for additional contextual factors that might influence the observed relationships.

Overall, this study sheds light on critical dynamics in expert-client interactions and highlights the adverse effect of client autonomy. It also provides valuable insights for designing effective communication and feedback mechanisms in various service industries.

Conflict of interest statement

There is no conflict of interest.

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A Additional Analysis

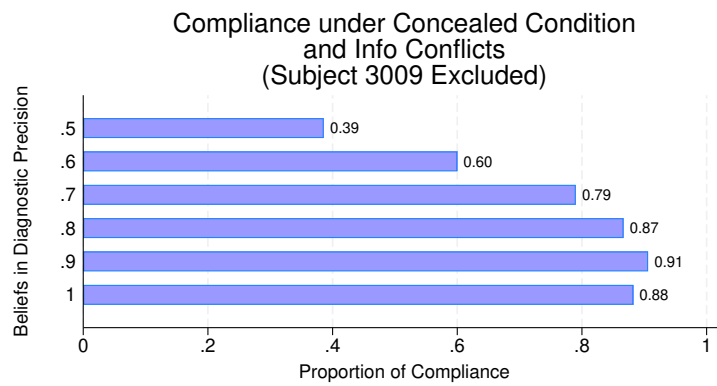


Figure A1: Client compliance with Subject 3009 Excluded.

Note: In this figure, the Subject with label 3009 is excluded, as a comparison to Figure 7(b).

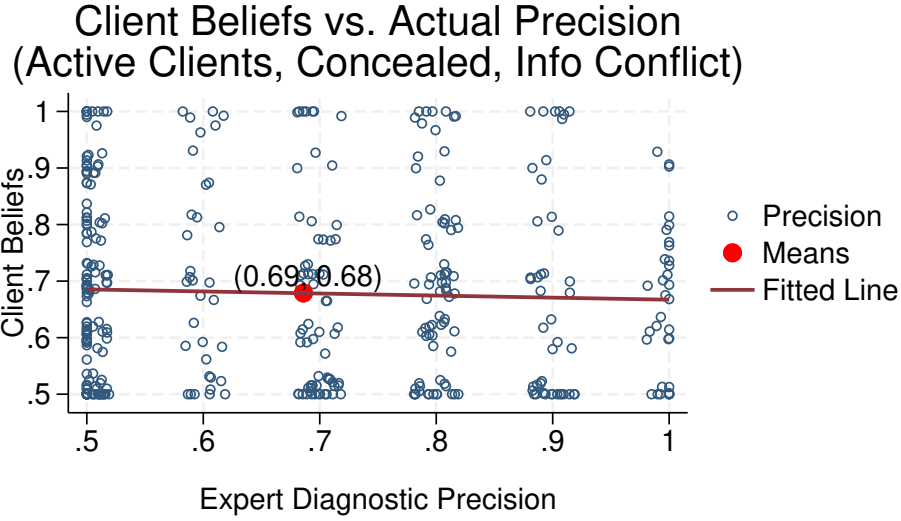


Figure A2: Jittered Scatter Plots of Clients' Beliefs and Actual Diagnostic Precision

Note: Data for this figure is restricted to the rounds in the Concealed condition with Active clients where their private signals conflict with experts' diagnostic results. The red dot indicates the mean of the diagnostic precision and the mean of clients' believed precision. The fitted line is derived from the regression of the actual precision to the believed precision, with the slope $\beta = -.036$ and $p\text{-value} = .465$.

Table A1: Logit Regression: Influence of Client Activeness on Probability of Solving a Problem

	DV: Indicator of Problem Solved	
	(1)	(2)
Concealed	0.551** (0.218)	0.544*** (0.210)
Active	-0.170 (0.167)	-0.270 (0.167)
Concealed $\times$ Active	-0.888*** (0.255)	-0.807*** (0.256)
Constant	0.975*** (0.131)	1.078*** (0.281)
Individual Controls	No	Yes
Round	No	Yes
Observations	2280	2280
Number of Individuals	114	114

Note: Individual controls include gender, ethnicity, whether the subject comes from Texas, subjects' altruism measured by choices in the dictator game, subjects' risk tolerance measured by choices in the investment game, and indicator of whether subjects major in economics or agricultural economics.

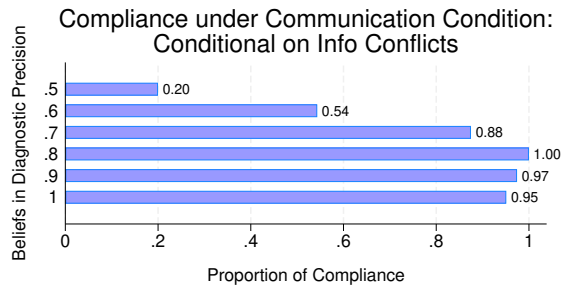
Standard error in parentheses, clustered at individual level. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table A2: Linear Regressions of Communication and Reputation Conditions on Diagnostic Precision

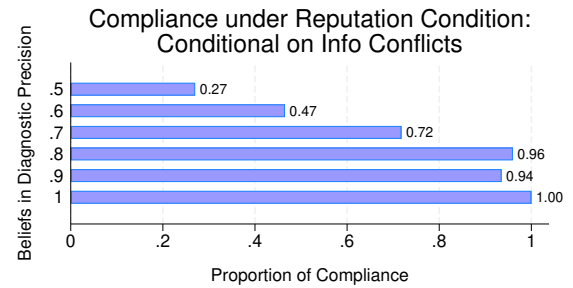
	DV: Diagnostic Precision	
	(1)	(2)
Alignment Parameter	0.301*** (4.27)	0.287*** (4.38)
Communication	-0.007 (-0.14)	-0.022 (-0.45)
Reputation	0.033 (0.67)	0.003 (0.07)
Communication $\times$ Alignment Parameter	0.014 (0.16)	0.023 (0.28)
Reputation $\times$ Alignment Parameter	-0.089 (-1.02)	-0.077 (-0.94)
Active	-0.043 (-1.42)	-0.048* (-1.75)
Active $\times$ Alignment Parameter	-0.110 (-1.57)	-0.101 (-1.57)
Communication $\times$ Active	-0.013 (-0.31)	-0.006 (-0.16)
Reputation $\times$ Active	-0.020 (-0.49)	-0.014 (-0.38)
Communication $\times$ Active $\times$ Alignment Parameter	0.133 (1.57)	0.114 (1.45)
Reputation $\times$ Active $\times$ Alignment Parameter	0.132 (1.62)	0.130* (1.75)
Constant	0.654*** (17.90)	0.744*** (13.31)
Individual Control	No	Yes
Round	No	Yes
Observations	3380	3380
N of Individuals	169	169

Note: Regressions in this table are restricted to the Concealed, Communication, and Reputation conditions, with the Concealed condition as the baseline. Both “Communication” and “Reputation” in the regression are binary indicators of treatment conditions. Individual controls include gender, ethnicity, whether the subject comes from Texas, subjects’ altruism measured by choices in the dictator game, subjects’ risk tolerance measured by choices in the investment game, indicator of whether subjects major in economics or agricultural economics.

Standard error in parentheses, clustered at an individual level. * $p < .1$; ** $p < .05$; *** $p < .01$.



(a) Communication Condition



(b) Reputation Condition

Figure A3: Proportion of Compliance Conditional on Information Conflicts

Note: Data is restricted to active clients only. “Compliance” means that the active client chooses the treatment consistent with the expert’s diagnostic result and treatment recommendation.



Figure A4: Word Cloud of Expert-Client Chat Messages in Communication Condition

Note: “Contract” is the framing of alignment parameter in the experiment; “accuracy” is the framing of diagnostic precision in the experiment.

B Institutional Changes: Communication and Reputation Conditions

This section reports the motivation, design, and findings of two institutional changes that are built upon the Concealed condition in the main text.

Under the Observable and Concealed conditions, clients remain unaware of any information about experts to maintain controlled testing environments. However, real-world interactions often involve clients having prior information about experts, aiding their decision-making. Therefore, I explore two institutional changes that are sometimes observed in service provisions: enhancing communication between clients and experts (Communication condition); establishing a rating system for experts to build reputations (Reputation condition). On one hand, clients may learn about experts' incentive alignment with them as well as their diagnostic precision through the conversation with experts; On the other hand, clients may share information on rating systems about experts that they interact with, where other clients can rely on this system to predict experts' diagnostic precision. Such information, although noisy, may change both the experts' and the clients' behaviors and therefore influence clients' probability of solving their problems. In this section, I go over the theoretical and empirical findings on communication and reputation systems in expert services to motivate these two treatments, and then present the treatment designs and results.

B.1 Communication and Reputation in Expert Services: Theoretical and Empirical Findings

Communication: Focusing particularly on the implications for healthcare, existing literature proposes that sufficient and effective communication is crucial for better healthcare service quality. There is a consensus within the scholarly community that a substantial number of patients and doctors experience obstacles related to inefficient or inadequate communication (El-Jawahri et al., 2019; Karras et al., 2003; Strull et al., 1984). Efficient and effective communication between patients and doctors is widely acknowledged as crucial for the quality of healthcare services, as highlighted in existing empirical studies. Alsan et al. (2019) conducted a field experiment randomly matching Black male patients to either White or Black doctors and found that Black male patients are more likely to accept the vaccine suggestion from Black doctors than White doctors. This result was attributed to better communication skills rather than just same-race matching, emphasizing the importance of communication in building trust and compliance. Moreover, previous studies have underscored the importance of patients' communication skills. Brekke et al. (2018) analyzed administrative data at the patient level in Norway and found that lower-educated patients received shorter consultation time and more medical tests, emphasizing the importance of patients' communication skills in influencing treatment decisions. Cegala et al. (2000) found that communication-skill training for patients significantly increased their compliance rates. In this study, I introduce an institutional change called the Communication condition to investigate the influence of expert-client communication on clients' problem-solving.

Reputation: While communication can be costly in terms of time and other resources, reputation, such as an online rating system, might offer a more affordable solution for improving problem-solving in expert services. Although from a theoretical perspective, if clients are unaware of product quality, their ratings of expert sellers may not be informative (Dulleck et al., 2011), empirical findings show that online rating systems provide useful information about experts. Ranard et al. (2016) compared Yelp reviews of hospital care with a traditional survey (Hospital Consumer Assessment of Healthcare Providers and Systems, HCAHPS) and found that Yelp reviews cover most HCAHPS topics and strongly predict hospital quality measures. Hence, online rating platforms can help patients gather information about health providers and enhance their trust in doctors. Kerschbamer et al. (2019) examined Yelp ratings and suggested that higher ratings are usually associated with lower prices. Despite this, online rating systems are still not widely used for gathering information about medical doctors compared to restaurants, cars, movies, or books (Hanauer et al., 2014). There is a lack of causal evidence measuring how online rating systems help consumers receive appropriate services and avoid being mistreated (Kerschbamer et al., 2019). In this study, I introduce a rating system in the Reputation condition allowing clients to share ratings on experts. As shown in Section 5, the Reputation condition effectively increases the probability for clients to solve their problems.

B.2 Experimental Design

The designs of the Communication and Reputation conditions are based on the Concealed condition – clients in those conditions do not observe experts’ diagnostic precision.

Communication: As shown in Figure ??, subjects enter a “negotiation” stage (highlighted in gray color) after experts choose diagnostic precision and before the diagnostic result is generated. In this stage, the matched client and expert have an opportunity to chat, and one should expect clients to persuade experts to increase the diagnostic precision level or to ask for information about the diagnostic precision.¹⁰ After the negotiation stage, experts have the opportunity to revise their diagnostic precision, then conduct the diagnosis and generate the treatment recommendation.

Reputation: Under the Reputation condition, there is a rating system for clients to rate their matched experts. As shown in Figure ??, at the beginning of each round, clients read their matched experts’ average ratings (on a scale of 1 to 5), and at the end of each round, clients give ratings to their matched experts, which will be included into the calculation of the expert’s average ratings for the subsequent rounds. After clients give ratings, experts are informed about the client’s new rating and their updated average rating. Hence, as the game continues, more and more ratings about each expert will be accumulated.¹¹

¹⁰In the first round, they chat for 3 minutes. The time is reduced to 60 seconds for subsequent rounds, since they learn to communicate efficiently. Based on pilot sessions, 60 seconds is enough for their conversation. In the instruction, subjects are informed that any information revealing their personal identity is prohibited in the chat.

¹¹In round 1, experts do not have any ratings. At the end of round 1, clients will give ratings to experts. Starting from round 2, clients will observe the actual rating of their matched experts.

Table B1: Proportion of Solved Problems for Clients

	Proportion of Solved Problems (%)		<i>p</i> -value: Active vs. Passive
	Active Clients	Passive Clients	
Concealed Condition	61.48	82.14	< 0.001
Communication Condition	72.25	75.00	0.337
<i>p</i> -value: Communication vs. Concealed	0.001	0.138	
Reputation Condition	70.74	78.89	<0.001
<i>p</i> -value: Reputation vs. Concealed	0.001	0.361	

Note: *p*-values for activeness comparison (the last column): linear regressions are performed using the indicator of the problem solved as the dependent variable and the indicator of the client being active as the independent variable, clustered at an individual level, with the *p*-values of the coefficient estimate reported. *p*-values for treatment condition comparison: linear regressions are performed using the indicator of the problem solved as the dependent variable and the indicator of treatment conditions as the independent variable, clustered at an individual level, with the *p*-values of the coefficient estimate reported.

B.3 Results and Discussions

B.3.1 Effects of Reputation and Communication on Problem Solving

Table B1 indicates that both the Reputation and Communication conditions significantly increase the proportion of solved problems for active clients. The positive impact of these two conditions on problem-solving is further supported by a logit regression using the indicator of a problem solved as the dependent variable. Table B2 reports the marginal effects of the Communication and Reputation condition from this regression, using the Concealed condition as the baseline. This table suggests that the Communication condition increases solved problems by about 10 percentage points, and the Reputation condition increases solved problems by around 9 percentage points. These results are robust after adding more controls. A coefficient test between these two treatment-condition indicators returns a *p*-value of 0.47, which suggests equal effectiveness of these two conditions in increasing the probability of solving a problem. In the next two subsections, I will examine the influence of these two institutional changes on the expert's and the client's behaviors.

B.3.2 Expert Diagnostic Precision under Institutional Changes

Do these two institutional changes improve active clients' problem-solving by increasing expert diagnostic precision? Panel A of Table B3 shows that compared to the Concealed condition, there seems lack of supporting evidence of higher diagnostic precision in the Communication or Reputation condition. Specifically, only for $\gamma = 0.2$ and $\gamma = 1$, the average diagnostic precision in the Communication condition is higher than in the Concealed condition at a 10% significance level.¹²

¹²Under the Communication condition, experts revise the diagnostic precision only 14% of the time, leading to an average increase of 14.4 percentage points in diagnostic precision (from 0.680 to 0.824).

Table B2: Logit Regression: Effects of Communication and Reputation Conditions on Solved Problems among Active Clients

	DV: Indicator of Problem Solved	
	(1)	(2)
Communication	0.108*** (0.030)	0.108*** (0.031)
Reputation	0.093*** (0.027)	0.086*** (0.030)
Individual Controls	No	Yes
Round	No	Yes
Observations	2360	2360
Number of Individuals	118	118

Note: Regressions in this table are restricted to data of Active clients under the Concealed, Communication, and Reputation conditions, with the Concealed condition as the baseline. Both “Communication” and “Reputation” in the regression are binary indicators that equal 1 if the individual of a certain round is under this corresponding treatment condition. Individual controls include gender, ethnicity, whether the subject comes from Texas, subjects’ altruism measured by choices in the dictator game, subjects’ risk tolerance measured by choices in the investment game, indicator of whether subjects major in economics or agricultural economics. Standard error in parentheses, clustered at individual level. * $p < .1$; ** $p < .05$; *** $p < .01$.

This finding is confirmed by a marginal effect analysis derived from a linear regression. This linear regression includes diagnostic precision as the outcome variable, and includes full factorials of the alignment parameter, the treatment condition indicators, and the client activeness indicator in the specification. The data of the regression is restricted to the Concealed, Communication, and Reputation conditions, with the Concealed condition as the baseline. See Table A2 in the Appendix for the raw regression results. For a clearer illustration of the treatment effects, Figure B1 shows the marginal effects of these two institutional changes on experts’ diagnostic precision, derived from Model (2) in Table A2 (the regression with individual controls). The left panel depicts the marginal effect of the institutional changes conditional on active clients, while the right panel focuses on passive clients. When interacting with active clients, we only observe a positive effect from the Communication condition on high-alignment experts (i.e., experts with $\gamma = 0.6$ and 1), and the Reputation condition does not increase diagnostic precision. Additionally, when interacting with passive clients, neither of these two institutional changes increases diagnostic precision.

As shown in earlier subsections, clients’ activeness discourages experts from investing more in improving diagnostic precision under the Concealed condition. To examine whether the Communication and Reputation conditions change experts’ differing attitudes toward active and passive clients, I further perform another marginal effect analysis on clients’ activeness derived from Model (2) in Table A2. Figure B2 visualizes the marginal effect of client activeness on diagnostic precision. Under both the Communication and Reputation conditions, client activeness does not reduce the diagnostic precision of experts with the highest

Table B3: Expert Average Diagnostic Precision

Panel A: Diagnostic Precision for Active Clients

	$\gamma = 0$	$\gamma = 0.2$	$\gamma = 0.6$	$\gamma = 1$	All
Concealed	0.599 (0.164)	0.642 (0.156)	0.756 (0.153)	0.771 (0.176)	0.696 (0.176)
Communication	0.563 (0.105)	0.734 (0.121)	0.784 (0.152)	0.923 (0.100)	0.727 (0.180)
<i>p</i> -value: Communication vs. Concealed	0.445	0.052	0.525	0.015	0.324
Reputation	0.613 (0.155)	0.676 (0.147)	0.813 (0.123)	0.833 (0.146)	0.715 (0.176)
<i>p</i> -value: Reputation vs. Concealed	0.786	0.496	0.211	0.330	0.536

Panel B: Diagnostic Precision for Passive Clients

	$\gamma = 0$	$\gamma = 0.2$	$\gamma = 0.6$	$\gamma = 1$	All
Concealed	0.607 (0.198)	0.730 (0.146)	0.875 (0.142)	0.894 (0.150)	0.787 (0.192)
Communication	0.613 (0.177)	0.759 (0.126)	0.869 (0.132)	0.921 (0.090)	0.760 (0.188)
<i>p</i> -value: Communication vs. Concealed	0.935	0.571	0.902	0.667	0.485
Reputation	0.661 (0.186)	0.765 (0.118)	0.870 (0.122)	0.873 (0.151)	0.784 (0.182)
<i>p</i> -value: Reputation vs. Concealed	0.415	0.416	0.916	0.761	0.924

Note: Standard deviations are in parentheses. *p*-values are from linear regressions using diagnostic precision as the outcome variable and the indicator of the two treatment conditions for comparison as the independent variable, clustered at the individual level. For each regression, I report the *p*-value associated with the estimated coefficient for the independent variable. These *p*-values indicate the statistical significance for the null hypothesis that the diagnostic precision is identical between the two conditions for comparison, accounting for repeated observations from each subject.

alignment parameter ($\gamma = 1$). This finding contrasts with the Concealed condition, where client activeness reduces those experts’ diagnostic precision by 0.146 percentage points. In addition, we do observe that for experts with $\gamma = 0.6$, clients’ active role significantly decreases experts’ diagnostic precision under the Communication and Reputation conditions. But point estimates of those marginal effects are only half compared to that under the Concealed condition. These results suggest that the Communication and Reputation conditions mitigate the negative impact of client activeness on expert diagnostic precision, particularly for high-alignment experts.

B.3.3 Client Active Participation under Institutional Changes

In this subsection, I illustrate how the Communication and Reputation conditions improve client problem-solving from the client side.

While the Communication condition increases high-alignment experts’ diagnostic precision, communication also enables clients to update their beliefs regarding experts’ diagnostic precision and incentive alignment (see Figure A4 in the Appendix for the word cloud of chat messages, where “incentive alignment” and “diagnostic precision” are frequently mentioned in subjects’ chats). Figure 3(a) depicts the distribution of clients’ beliefs about diagnostic precision alongside the actual precision. The fitted line of the beliefs and actual precision is upward-sloping with a statistically significant coefficient estimation ($\beta = .643$, *p*-value = .000). This finding suggests that compared to the noisy belief in the Concealed condition (Figure A2 in Appendix), under the Communication condition, clients’ beliefs on the diag-

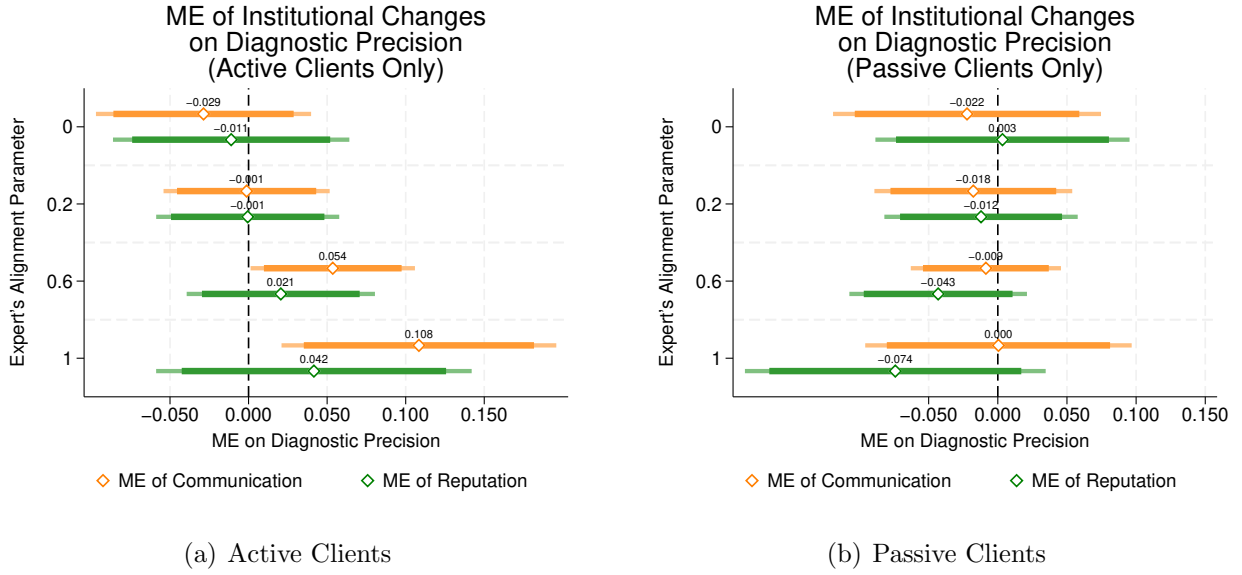


Figure B1: Marginal Effect of Institutional Changes on Diagnostic Precision

Note: Data is restricted to the Concealed, Communication, and Reputation conditions, using the Concealed condition as the baseline. These two figures illustrate the marginal effect analysis derived from Column (2) of Table A2 in the Appendix.

nostic precision become more accurate. Given that clients are more compliant when they believe in higher diagnostic precision (Figure A3(a) in Appendix), clients under the Communication condition can make more informed choices by following the highly precise medical treatment recommendations and disregarding imprecise ones.

In contrast, active clients under the Reputation condition are not able to correctly guess experts' diagnostic precision, as shown in Figure 3(b). Furthermore, the previous subsection also shows that experts do not increase the investment in active clients under the Reputation condition compared to the Concealed condition.

Figure B4 sheds light on why active clients under the Reputation condition still manage to increase problem-solving: they effectively utilize the rating system to share information about experts' diagnostic precision. In the left panel Figure 4(a), clients' observed expert average ratings are divided into nine equal-sized bins, and within each bin, the proportion of clients deciding to comply with their matched experts is plotted. This scatter plot, together with an upward-sloping fitted line, indicates that when facing an information conflict, clients are more inclined to follow experts who are of higher average ratings.

Moving to the right panel Figure 4(b), I further examine whether the average rating on experts conveys useful information about experts' actual diagnostic precision. Recall that after each interaction with experts, clients assign a rating to their experts, on a scale of 1 to 5 (5 the best). In Figure 4(b), for each rating level that clients give, I calculate the corresponding average of the diagnostic precision from those rated experts. This panel demonstrates a positive association between ratings and diagnostic precision, indicating that clients assign higher ratings to experts with higher diagnostic precision. Therefore, clients'

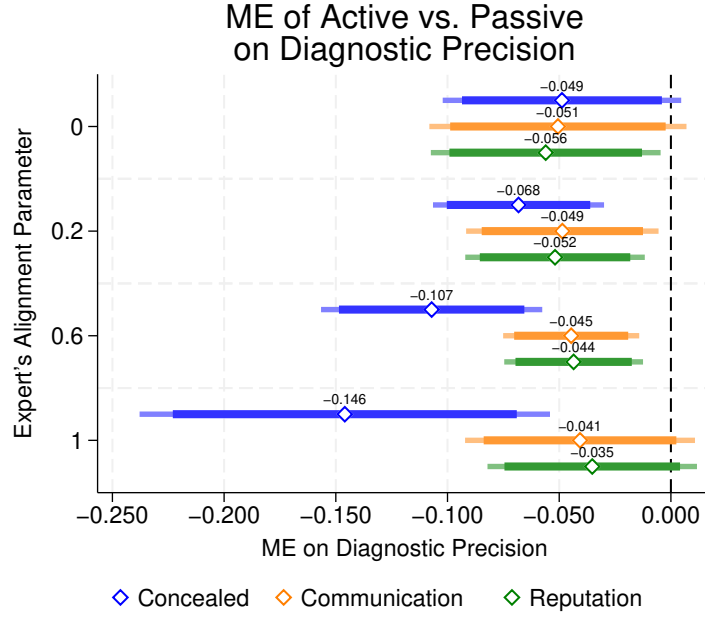
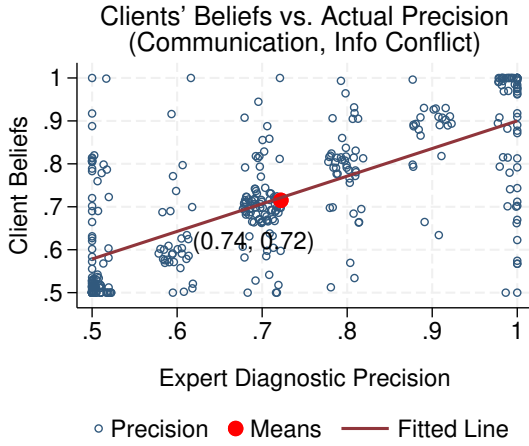


Figure B2: Marginal Effect of Client Activeness on Diagnostic Precision under Institutional Changes

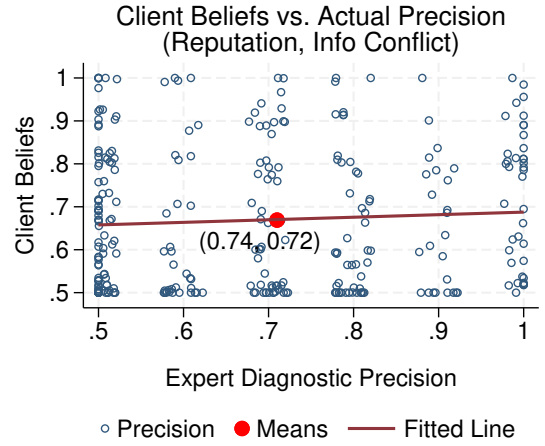
Note: This figure is a marginal effect analysis derived from Model (2) in Table A2. The width of bars are the 95% and 90% confidence intervals.

ratings effectively convey valuable information about experts' diagnostic precision. Tying this finding to the left panel, these two panels suggest that clients adeptly utilize the rating system by sharing information about experts to make informed decisions.

In summary, compared with the Concealed condition, both the Reputation and Communication conditions improve client problem-solving. Under the Communication condition, clients benefit from the ability to form more accurate beliefs about experts' diagnostic precision through conversations. This enables them to make informed decisions by following the recommendations of highly precise experts and disregarding those who are less precise. In contrast, the Reputation condition provides information about the experts in a less direct but still effective way. Rather than directly learning about experts' diagnostic precision or incentive alignment, clients rely on a rating system to share and utilize valuable insights about their matched experts. Clients' ratings serve as an indicator of diagnostic precision, allowing clients to follow recommendations from experts with higher ratings, thereby improving their problem-solving.



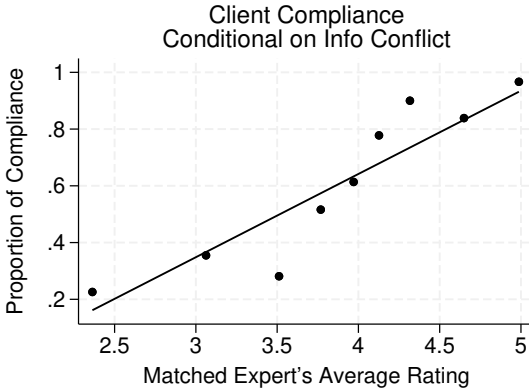
(a) Communication Condition



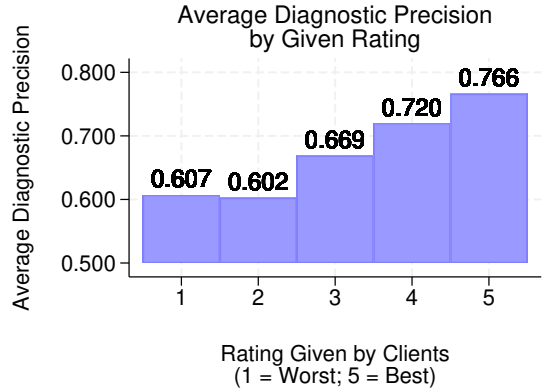
(b) Reputation Condition

Figure B3: Scatter Plots of Clients' Beliefs on Diagnostic Precision and Actual Diagnostic Precision

Note: Data for these two panels are restricted to rounds with Active clients where their private signals conflict with experts' diagnostic results. In both left- and right panels, the red dot indicates the mean of the diagnostic precision and the mean of clients' believed precision; the fitted line is derived from the regression of the actual precision to the believed precision. On the left panel (Communication condition), the slope of the fitted line is $\beta = .643$ and $p\text{-value} = .000$. On the right panel (Reputation condition), the slope of the fitted line is $\beta = .060$ and $p\text{-value} = .285$.



(a) Active clients' compliance by experts' average rating conditional on info conflict.



(b) Average diagnostic precision by ratings given from clients after each round of interaction, conditional on information conflict.

Figure B4: Client Ratings, Expert Diagnostic Precision, and Client Compliance

Note: The left panel illustrates the relationship between experts' average rating and clients' compliance. The expert ratings that clients observe at the beginning of a round are divided into nine equal-sized bins. Within each bin, I plot the proportion of active clients who comply with those experts when there are information conflicts. The right panel illustrates the relationship between ratings given to experts and experts' diagnostic precision, conditional on information conflict.

C Pre-Registration File

(See the next page for the full pre-registration.)

CONFIDENTIAL - FOR PEER-REVIEW ONLY**An Theoretical and Experimental Investigation of Patient-Physician Relationship (#105234)**

Created: 08/22/2022 07:02 PM (PT)

This is an anonymized copy (without author names) of the pre-registration. It was created by the author(s) to use during peer-review.
A non-anonymized version (containing author names) should be made available by the authors when the work it supports is made public.

1) Have any data been collected for this study already?

It's complicated. We have already collected some data but explain in Question 8 why readers may consider this a valid pre-registration nevertheless.

2) What's the main question being asked or hypothesis being tested in this study?

We use a laboratory experiment to analyze the interaction between patients and doctors. We are verifying the following predictions:

1. Compared with a passive patient who always follows the doctor's recommendations, the doctor is more likely to choose lower precision levels when interacting with an active patient;
2. The active patient's noncompliance may occur only when his/her private information about the disease conflicts with the doctor's diagnostic result. And when there exists such information conflict, the patient will go against the doctor when he/she observes/believes that the doctor's diagnostic precision is lower than the accuracy of her private information;
3. Compared with the case when the diagnostic precision is observable, when the diagnostic precision is not observable, patients are more likely to go against doctors, and doctors are choosing lower diagnostic precision levels on average.

And we ask the following empirical questions:

1. Does communication/reputation increase patients' compliance?
2. Does communication/reputation encourage doctors to invest more to increase diagnostic precision?
3. Does communication/reputation improve patients' health outcome?

3) Describe the key dependent variable(s) specifying how they will be measured.

- 1) Patient's willingness to pay for being an active patient;
- 2) Patient's compliance with the doctor's treatment recommendation;
- 3) Doctor's choices in diagnostic precision;
- 4) Medical treatment outcome.

4) How many and which conditions will participants be assigned to?

We will implement four treatment variations of a doctor-patient game. In this game, a patient is trying to figure out an unknown disease and receive desired medical treatment, and a doctor will invest some effort to generate a diagnostic result and recommend a medical treatment for the patient.

Treatment conditions vary in the followings:

- T1: Doctors' diagnostic precision is observable to patients.
T2: Doctors' diagnostic precision is not observable to patients.
T3: Doctors' diagnostic precision is not observable to patients, and patients and doctors will directly chat with each other during the game.
T4: Doctors' diagnostic precision is not observable to patients, and patients will read the average ratings of a doctor at the beginning of a medical visit and will rate their matched doctors after each medical visit.

5) Specify exactly which analyses you will conduct to examine the main question/hypothesis.

Regression analysis on the primary dependent variables, as a function of the treatment conditions. The analysis of patient's behavior will include the control for patient's risk preference. The analysis of doctor's behavior will include the control for doctor's altruism level.

6) Describe exactly how outliers will be defined and handled, and your precise rule(s) for excluding observations.

We do not plan to exclude any observations, absent technical issues during the data collection.

7) How many observations will be collected or what will determine sample size? No need to justify decision, but be precise about exactly how the number will be determined.

We aim to have a minimum of 74 subjects per treatment, for a total of at least 296 participants.

8) Anything else you would like to pre-register? (e.g., secondary analyses, variables collected for exploratory purposes, unusual analyses planned?)

We will include additional experimental tasks and surveys to measure participants' risk preferences and altruism and collect participants' demographic information (age; gender; major).

Existing data: We have conducted a few preliminary sessions to pilot the implementation of the game in the lab and calibrate the length of a session.

D Lab Part Instructions

Introduction¹³

Thank you for participating in the experiment!

Please do not communicate with other participants during the experiment. If you have any questions, please raise your hand, and an experimenter will come to assist you. If you violate this rule, we may have to exclude you from the experiment and from all payments.

This experiment contains two parts. The first part is a decision-making game, which will repeat for 20 rounds, and at the end of the experiment, **one out of 20 rounds will be randomly selected and paid**. The second part is a questionnaire.

You will receive a \$10 show up fee for your participation. This will be yours to keep. You also will have an opportunity to earn more, based on your decisions, the decisions of others, and luck. So, please read the following instructions carefully.

The experiment will use the currency named “tokens”, with the following exchange rate:

$$\begin{aligned} 1 \text{ token} &= 10 \text{ cents} \\ (100 \text{ tokens} &= 10 \text{ dollars}) \end{aligned}$$

At the end of the experiment, your earnings in tokens will be converted into dollars. So, the more tokens you earn from the experiment, the more money you will be able to make. Your final payment will consist of three parts: (1) \$10 show up fee; (2) your earnings from the online survey that you have already completed; (3) the earnings from today’s experiment.

Instruction

In this experiment, there are two roles in a game: **client** and **expert**. You will be randomly assigned to one of the two roles, and will remain the same role for the whole experiment. There will be 20 rounds of games, and in each round, the matching of client and expert is random – in each round you should expect interacting with a different person.

Generally speaking, this game is about solving the client’s problem. For each round, the client is facing a problem. The problem will always be either Problem A or Problem B, with equal probability (50%). This means that for different rounds, the client can have different problems. Each problem has its own correct solution: Problem A can be solved by Solution A*; Problem B can be solved by Solution B*.

The client does not know which problem he/she is facing, nor the correct solution to the problem. So, he/she has to consult an expert to identify the problem. The expert is randomly matched to him/her by the computer. The expert conducts a costly “diagnosis” to try to identify the problem and provide a recommendation of a solution to the client. If the client receives the correct solution to the problem, the problem will be successfully solved.

Next, you will read the detailed description of the client’s and the expert’s roles.

¹³This document combines the instructions of the four treatment conditions.

Client's Role

Suppose you have been assigned as a client.

At the beginning of each round, you will be facing either Problem A or Problem B, and your goal is to solve your problem. If you receive the correct solution (“Solution A*” for having “Problem A”; “Solution B*” for having “Problem B”), you will get **120 tokens** for this round. If you receive the incorrect solution (“Solution B*” for having “Problem A”; “Solution A*” for having “Problem B”), you will get **20 tokens** for this round.

You do not know exactly which problem it is. Instead, you will receive a signal about the problem. The signal will be either “Problem A” or “Problem B”, appearing on the computer screen. You should be aware that this signal may NOT be true: it only correctly identifies your problem **60%** of the time.

For example, if you see that the signal on the screen is “Problem A”, you should be aware that there is a 60% chance that you actually have “Problem A”, and 40% chance that you have “Problem B”.

Another information source for you to guess your problem is the expert. You are randomly matched with an expert for each round. Your expert will conduct a “diagnosis” to identify your problem and recommend a solution to you. The way that the expert diagnoses your problem is by choosing the “diagnostic accuracy”, i.e., how accurately he/she can identify your problem, ranging from 50% to 100%. 50% means that the expert’s diagnosis will be correct only 50% of the time. And 100% means that the expert’s diagnosis will be correct 100% of the time.

For example, suppose that the problem you are actually facing is “Problem A”. If the expert chooses 50% as the diagnostic accuracy, then the expert’s diagnostic result will be “Problem A” for only 50% of the time, and for the other 50% of time, the diagnostic result will be “Problem B”. If the expert chooses 100%, then the expert’s diagnosis will be “Problem A” for 100% of the time, which means that the expert will always be correct. (The expert’s decision process will be explained later).

[Observable condition:] You will be able to observe the diagnostic accuracy level chosen by the expert.

[Concealed, Communication, and Reputation conditions:] You will NOT be able to observe the diagnostic accuracy level chosen by the expert.

After diagnosing your problem, the expert will recommend a solution to you with the following rule: if the expert’s diagnosis is “Problem A”, the expert always recommends “Solution A*”; if the expert’s diagnosis is “Problem B”, the expert always recommends “Solution B*”.

As a client, you will be randomly allocated to be “Passive” or “Active”. If you are Passive, you must follow the expert’s recommended solution. If you are Active, you will freely choose the actual solution after seeing the expert’s recommended solution. In other words, if you are “Passive”, then your matched expert will choose a solution for you; if you are “Active”, you will choose the solution yourself.

Additionally, if you are assigned to be Passive, we will ask you to predict whether your matched expert will solve your problem. Your correct prediction in a round will bring you extra 10 tokens of reward, if that certain round is randomly chosen for payment at the end of the experiment.

You will have a 30% chance to be assigned as a “Passive” client, and 70% chance to be assigned as an “Active” client. At the beginning of the experiment, you will be informed about the “Passive” and “Active” assignment, and this status will remain the same for the whole experiment.

For each round, your income is:

$$\text{Your income in this round} = \begin{cases} 120 \text{ tokens} & \text{if problem is solved} \\ 20 \text{ tokens} & \text{if problem is not solved} \end{cases}$$

Expert’s Role

Now suppose that you are the expert.

You will be matched with a client for each round. Your matched client does not know which problem (Problem A / Problem B) he/she has, and you have the ability to diagnose the problem. Your diagnosis of the problem will also determine a recommendation of a solution for the client (Solution A* or Solution B*).

To diagnose the client’s problem, you will first choose your diagnostic accuracy, which determines the probability that you correctly identify the client’s problem. The accuracy level is associated with costs in tokens, according to the following cost table. You should notice that the higher the diagnostic accuracy you choose, the more you have to pay for this diagnosis.

Diagnostic accuracy	50%	60%	70%	80%	90%	100%
Cost of diagnosis in tokens	0	1	4	9	16	25

After choosing the accuracy level, the computer will generate a diagnosis of either “Problem A” or “Problem B” with your chosen accuracy. As already explained, the higher the accuracy level you choose, the more likely you will be able to correctly identify the client’s problem. Your diagnosis will automatically generate a corresponding recommendation and be sent to your client (Solution A* for Problem A; Solution B* for Problem B).

However, some clients might not necessarily follow your recommendation. If the client is “Active”, then he/she will have an opportunity to freely choose the solution after seeing your diagnosis and recommendation, which means that he/she may go against your recommendation. If the client is “Passive”, he/she will always follow your recommendation. **For each round, we will inform you your matched client’s type (“Active” / “Passive”) before you choosing your diagnostic accuracy.**

For each round, you will receive **80** tokens as your base income.

Additionally, as an expert, you will be randomly assigned to one of the four contracts: 0-Contract, 20-Contract, 60-Contract, or 100-Contract. These contracts determine the extra income you will earn from successfully solving the client’s problem. The following table summarizes the extra income from each type of contract:

Contract	Your Extra Income If Problem Solved
0-Contract	0 Tokens
20-Contract	20 Tokens
60-Contract	60 Tokens
100-Contract	100 Tokens

At the beginning of the experiment, you will be informed about the assigned contract, i.e., the extra tokens you can earn from solving the client’s problem, and this will remain the same for the whole experiment. Your matched client will NOT be informed about your contract, i.e., the extra income that you are earning from solving his/her problem.

For each round, your income is:

$$\text{Income for one round} = 80 \text{ tokens} + \text{Extra Income from Problem Solving} - \text{Cost of diagnosis}$$

[Communication condition:]
Negotiation Stage

During each round, after the expert chooses the diagnostic accuracy and before the expert conducts the diagnosis, the expert and client will have a chance to negotiate through a chat box on the computer screen.

While chatting, you should NOT share any information that identifies (or helps to identify) who you are. You should not share your background information, for example, name, age, gender, location in the lab, etc. If you break this rule, we will exclude you from any earnings from the experiment.

For the first round, the negotiation stage will last for 3 minutes (180 seconds). After the first round, this negotiation will last for 1 minute (60 seconds). There will be a countdown on the top of the screen that you can refer to. Once the time is out, you will move on to the next stage of the experiment. And if you and your matched player finish your negotiation ahead of time, you both still have to wait until time is out.

After the negotiation stage, the expert will decide whether or not to revise his/her chosen diagnostic accuracy level. After that, the expert will actually conduct the diagnosis and generate the recommendation of solution for the client.

[Reputation condition:]
Rating System

There is an additional element in this game: a rating system where clients can read and give ratings to their matched experts.

At the beginning of each round, the client will first receive the matched expert’s average rating, on a scale of one to five (one the worst, five the best), given by other clients that have interacted with this expert.

At the end of each round, after the client receiving the feedback on the problem solving, i.e., whether he/she successfully solves the problem, the client will also need to give a rating

to the matched expert, on a scale of one to five. This rating will be included in the average rating of this matched expert, and other clients who are matched with this expert in the next round will see this updated average rating.

At the end of each round, the expert will be informed about the rating given by the matched client and the updated overall average rating. The rating does not change the expert's payoff.

You should be aware that in the first round, there will be no average ratings of the expert provided for the client. This is because in round 1, none of the clients have given ratings to their matched experts. After the first round, all experts start to have their average ratings that their future matched clients can read.

Instructions, Continued

Now, the computer will randomly assign your roles. Notice that once you are chosen as a client or an expert, you will remain in the same role for the whole experiment.

After receiving your role assignment, you will go through a review about the procedure for your role, and then answer several questions to make sure that you understand the game rules.

Again, your decisions will affect how much you and your matched player can earn from the experiment, so, please pay attention to the instructions. If you have any questions, please raise your hand, and an experimenter will come to assist you.

E Online Part Instructions and Screenshots

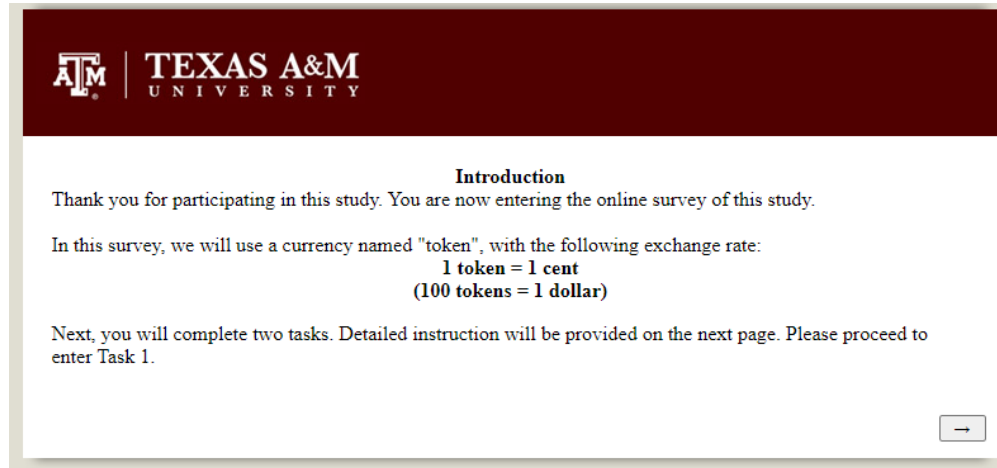


Figure E1: Online Survey: Introduction

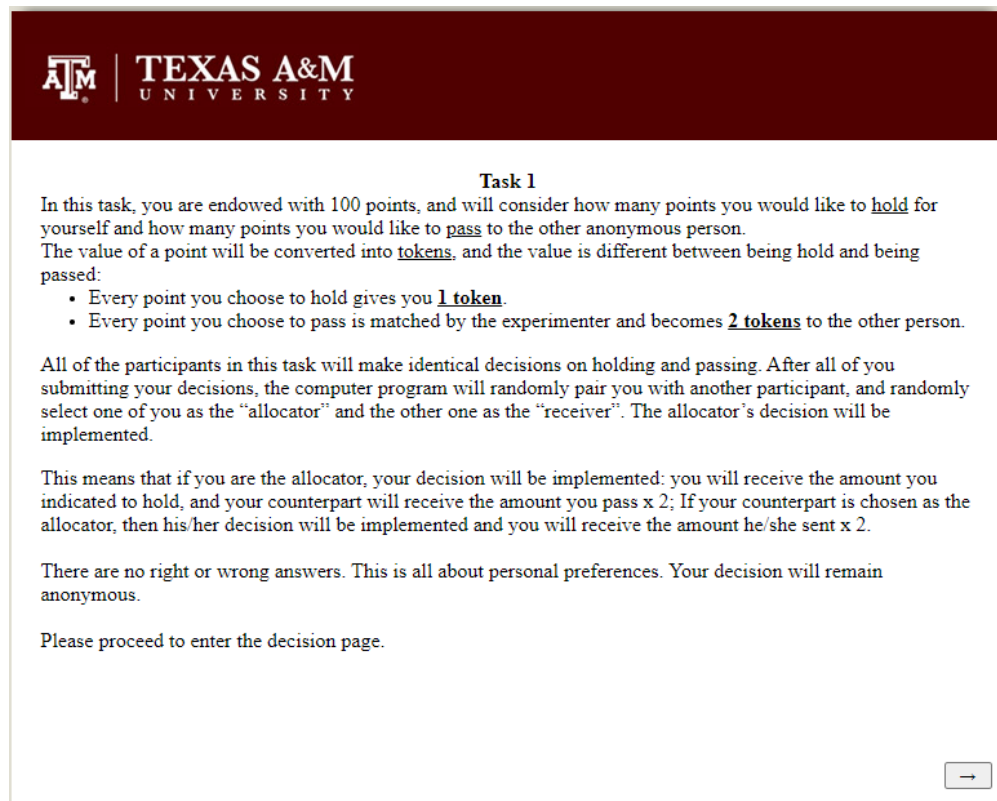


Figure E2: Online Survey: Dictator Game – Instruction



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Now please determine the allocation of 100 points between you and your matched anonymous opponent. Again, notice that each point brings about different amount of tokens to you and your opponent.

Holding points	Tokens that you will earn	Passing points	Tokens that your opponents will earn
☐ 100	100	0	0
☐ 90	90	10	20
☐ 80	80	20	40
☐ 70	70	30	60
☐ 60	60	40	80
☐ 50	50	50	100
☐ 40	40	60	120
☐ 30	30	70	140
☐ 20	20	80	160
☐ 10	10	90	180
☐ 0	0	100	200

→

Figure E3: Online Survey: Dictator Game – Decision Screen



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Task 2

Task 2 is a simple question. Please answer the following question by dragging the slider:

Do you generally donate a lot of time or money to help others, or do you focus primarily on taking care of yourself and your family?
(0 = "Donate a lot to others"; 10 = "Focus on self and family")

Donate a lot to others
0
1
2
3
4
5
6
7
8
9
10
Focus on self and family

Please drag the slider to indicate your answer.

→

Figure E4: Online Survey: Survey Elicitation of Altruism

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Task 3

You are endowed with 100 tokens and you need to choose the portion of this amount (between 0 and 100 tokens, inclusive) that you wish to invest in a risky option. Those tokens not invested are yours to keep.

The risky investment: There is a 50% chance that the investment in the risky asset will be successful. If it is successful, you receive 3 times the amount you chose to invest; if the investment is not successful, you lose the amount invested.

How Do We Determine if the Investment is Successful? The computer will randomly select a number between 1 and 10. If the number is 1, 2, 3, 4, or 5, your investment is not successful. If the number is 6, 7, 8, 9, and 10, your investment is successful.

You need to make a decision about the number of tokens that you wish to invest. In the test box please indicate the number of tokens you wish to invest. If your investment is successful, you will earn three times the amount invested; if it is unsuccessful, you will lose your investment.

Now, please insert the tokens that you would like to invest. You can type in any whole number between 0 and 100.

Number of tokens to invest:



Figure E5: Online Survey: Investment Game

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You are almost done with the online survey. To finalize this online survey, we also would like to collect your TAMU email address. We collect your email address in order to match your earnings and data from this online survey with your earnings and data from the in-person experiment.

Your contact information will remain confidential and only IRB-approved persons have access to it. Your contact information will also be stored separately from your data, so that no one can identify you from the data.

Now please insert your TAMU email address.

Figure E6: Online Survey: Email Collection for Payment

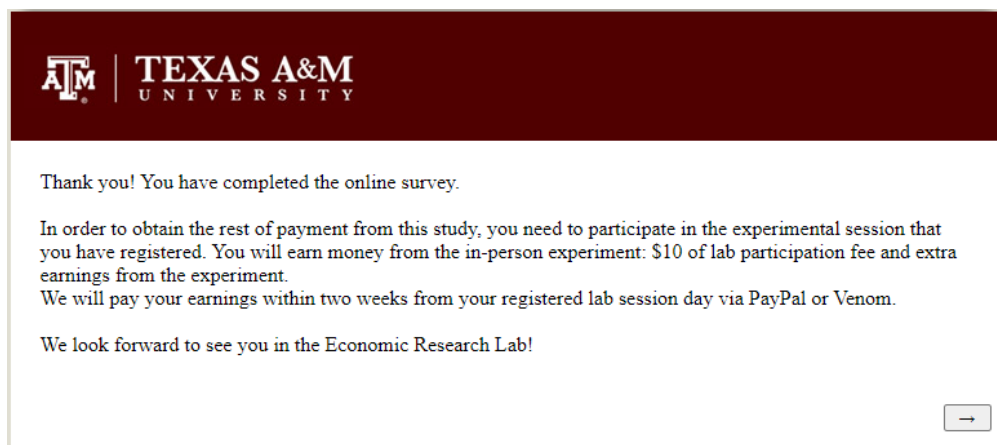


Figure E7: Online Survey: End of Task